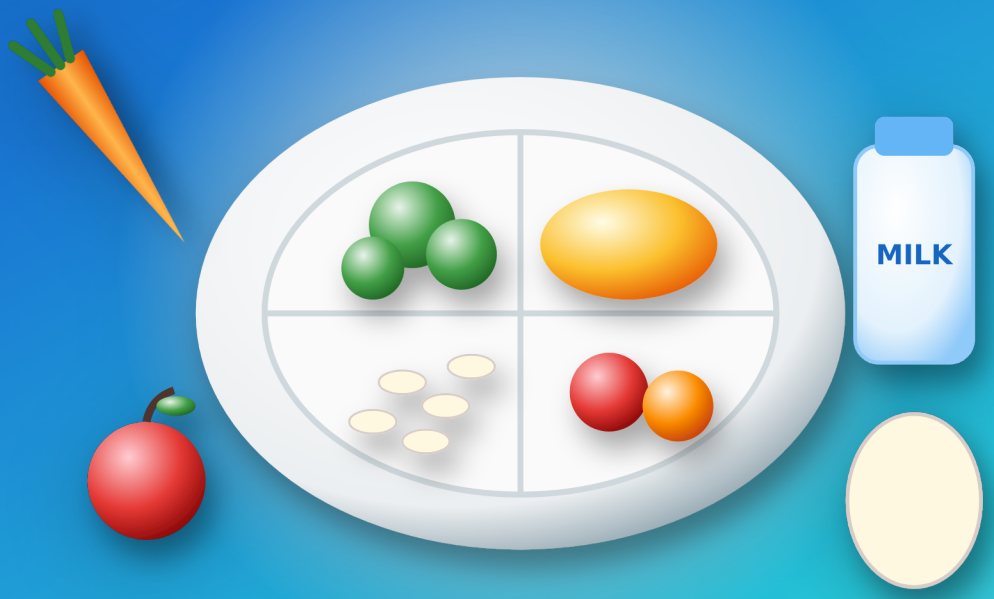
GSSSB · STAFF NURSE EXAM

Nutrition & Dietetics for Nurses

Complete Colourful Notes · 3D Diagrams · Mnemonics · Exam Points



10 Chapters | 3D Coloured Diagrams | Nutrients · RDA (ICMR-NIN 2020) | Deficiency diseases · Malnutrition | Therapeutic diets · Programmes | Exam Points (PYQ style)

Hinglish explanation + English exam terms



Sister Asha
Aapki Nursing teacher

Index – Kya padhenge?

1 **Introduction, Food Groups, Energy & Balanced Diet**
Definitions, functions of food, classification of foods & nutrients, ICMR-NIN food groups & My Plate, balanced diet, energy – calorie, BMR, SDA, energy requirements, RDA/ EAR (ICMR-NIN 2020), dietary guidelines 2024, reference man & woman, cooking methods

2 **Macronutrients – Carbohydrates, Proteins, Fats, Fibre & Water**
Classification, functions, sources, requirements, digestion & deficiency/ excess of carbohydrates, glycaemic index, dietary fibre, proteins – amino acids, protein quality (BV, NPU, PER, chemical score), fats – fatty acids, EFA, cholesterol, lipoproteins, trans fats, water & electrolytes

3 **Vitamins**
Classification, fat-soluble vitamins A, D, E, K – functions, sources, RDA, deficiency, toxicity & prophylaxis; water-soluble B-complex (B1, B2, B3, B5, B6, B7, B9, B12) & vitamin C; antioxidants; national prophylaxis

4 **Minerals & Trace Elements**
Macro & micro minerals; calcium & phosphorus, iron (absorption, anaemia, IFA prophylaxis, Anaemia Mukt Bharat), iodine & IDD, zinc, fluoride, magnesium, sodium & potassium, selenium, copper, chromium, manganese – functions, sources, RDA & deficiency

5 **Nutrition Through the Life Span**
Nutrition in pregnancy & lactation, breastfeeding – composition of breast milk, colostrum, advantages, technique, problems, expressed breast milk; infant feeding, complementary feeding (IYCF), weaning foods, preschool & school children, adolescents, adults & elderly; low birth weight & preterm feeding

6 **Malnutrition & Nutritional Assessment**
Forms of malnutrition, PEM – marasmus vs kwashiorkor, SAM & MAM, NRC & CMAM, F-75/ F-100/ RUTF, refeeding syndrome; stunting, wasting, underweight; classifications (Gomez, IAP, Waterlow, WHO); obesity & BMI; nutritional assessment – ABCD, anthropometry, growth charts, MUAC, skinfold; NFHS-5 data

7 **Therapeutic Nutrition & Diet in Diseases**
Principles of diet therapy, hospital diets (clear liquid, full liquid, soft, regular), modified diets; diet in diabetes, hypertension & CVD, renal disease, liver disease, GI disorders, peptic ulcer, diarrhoea, constipation, fever & infections, TB, burns, surgery, cancer, gout, anaemia, PKU & inborn errors; enteral (tube) feeding & parenteral nutrition; drug–food interactions; nurse's role

8 **Food Hygiene, Safety, Preservation & Adulteration**
Food hygiene & handling, milk hygiene & pasteurisation, food-borne diseases & food poisoning, food preservation methods, food additives, food fortification (+F), food adulteration & simple tests, FSSAI & FSS Act 2006, food labelling, toxins in food

9 **National Nutrition Programmes & Nutrition Education**
ICDS & Saksham Anganwadi, POSHAN Abhiyaan/ Poshan 2.0, PM POSHAN (mid-day meal), Anaemia Mukt Bharat, Vitamin A prophylaxis, NIDDCP, NRCs, MAA, PMMVY, PDS & NFSA, Eat Right India, Poshan Maah, SDGs & global targets, nutrition organisations, nutrition education & nurse's role

10 **Quick Revision – Master Tables & 150 One-Liners**
Richest sources table, deficiency diseases table, RDA quick table, normal values, important days & programmes, 150 rapid-fire one-liners and exam strategy

★ **Key Points** – sabse important facts

Asha Didi ki Tip – samjhane wali baat

! Nursing Alert – galti mat karna

✎ **Exam Point** – PYQ style, baar baar aata hai

♫ **Mnemonic** – yaad karne ki trick

D Definition – exact definition

1 Introduction, Food Groups, Energy & Balanced Diet

1.1 Basic Terms

D Definitions

- **Nutrition** – science of food, the nutrients in it, their **action, interaction & balance in relation to health and disease**, and the processes of ingestion, digestion, absorption, transport, utilisation & excretion.
- **Food** – anything eaten/ drunk that nourishes the body; **nutrients** – chemical substances in food needed for growth, maintenance & energy.
- **Dietetics** – application of nutrition science to feeding individuals/ groups in health & disease; **dietitian** plans therapeutic diets.
- **Diet** – food & drink regularly consumed; **balanced diet** – diet containing **all nutrients in correct amounts & proportions** to meet requirements, with a **small margin of safety** for lean periods.
- **Malnutrition** – pathological state from **relative or absolute deficiency or excess** of one or more nutrients – **undernutrition, overnutrition, specific deficiency, imbalance** ("double/ triple burden").
- **Optimal nutrition** – best possible nutritional status for health.
- **Father of nutrition – Antoine Lavoisier** (respiration & calorimetry); **Father of Indian nutrition research – Robert McCarrison** (Nutrition Research Laboratories, Coonoor 1918 → **NIN Hyderabad**, 1958).

1.2 Classification of Nutrients & Foods

- **Macronutrients** (required in **grams**) – **carbohydrates, proteins, fats** (+ water); **micronutrients** (mg/ µg) – **vitamins & minerals**.
- **By function** – **energy-yielding** (carbohydrates, fats, proteins), **body-building** (proteins – milk, meat, pulses, eggs), **protective/ regulatory** (vitamins, minerals – fruits, vegetables, milk, eggs).
- **Water** – most essential nutrient (death in days without it); ~**60 %** of body weight.
- **Dietary fibre** – non-digestible plant polysaccharides – essential for bowel health.

ICMR-NIN food groups

Group	Main nutrients
Cereals & millets (rice, wheat, jowar, bajra, ragi)	**Energy (CHO)**, protein (lysine-poor), B-vitamins, fibre; **ragi – richest in calcium**; bajra – iron
Pulses & legumes (dals, soybean, chana)	**Protein (methionine-poor, lysine-rich)**, B-vitamins, iron, fibre; **soybean – richest plant protein (~40 %)**
Milk & milk products, meat, fish, eggs	**High-quality protein**, calcium (milk), B12 (animal only), iron (meat), vitamin A

Group	Main nutrients
Vegetables (green leafy, roots & tubers, others) & **fruits**	**Vitamins (A – carotene, C, folate), minerals (iron, calcium), fibre, antioxidants**
Nuts & oilseeds, fats & oils, sugars	**Energy (concentrated)**, essential fatty acids, vitamin E

- **Basic 5 food groups** (old) – cereals; pulses; milk & meat products; fruits & vegetables; fats & sugars.
- **Cereal + pulse combination (ideal 3:1 or 4:1, e.g. khichdi, idli, dal-rice)** – **mutual supplementation** of amino acids (cereal lacks **lysine**, pulse lacks **methionine**) → improves protein quality.

1.3 Energy

- **Unit – kilocalorie (kcal)** – heat required to raise **1 kg of water by 1 °C** (from 14.5 to 15.5 °C); **1 kcal = 4.184 kJ** (≈ 4.2 kJ); SI unit – **joule** (megajoule).
- **Physiological fuel values (Atwater factors)** – **carbohydrate 4 kcal/g, protein 4 kcal/g, fat 9 kcal/g, alcohol 7 kcal/g**; fibre ~2 kcal/g.
- **Bomb calorimeter** – measures gross energy of food.
- **Total energy expenditure = BMR (~60–70 %) + physical activity (~20–30 %) + thermic effect of food/ SDA (~10 %)** (+ growth, pregnancy, lactation).

Basal metabolic rate (BMR)

- Energy needed to maintain **vital functions at complete physical & mental rest**, **12–14 hrs after last meal (post-absorptive)**, awake, lying, at **comfortable temperature (thermoneutral)**.
- **~1 kcal/kg/hr (adult male)**; ~**1500–1700 kcal/day** adult man.
- **↑ BMR** – **infancy & childhood (highest per kg)**, growth, pregnancy (later), **fever (~13 % per °C rise / 7 % per °F)**, **hyperthyroidism**, cold climate, stress/ adrenaline, **male (more lean mass)**, athletes, burns.
- **↓ BMR** – **old age, starvation/ malnutrition, hypothyroidism, sleep (~10 % lower), female**, obesity (per kg), sedentary.
- **SDA (specific dynamic action)/ thermic effect of food** – extra heat produced by digestion & metabolism – **protein ~30 % (highest)**, carbohydrate ~5–6 %, fat ~4 %; mixed diet ~10 %.
- **Physical activity level (PAL)** – sedentary (~1.53), moderate (~1.8), heavy (~2.3) × BMR.

1.4 Nutrient Requirements (ICMR-NIN 2020)

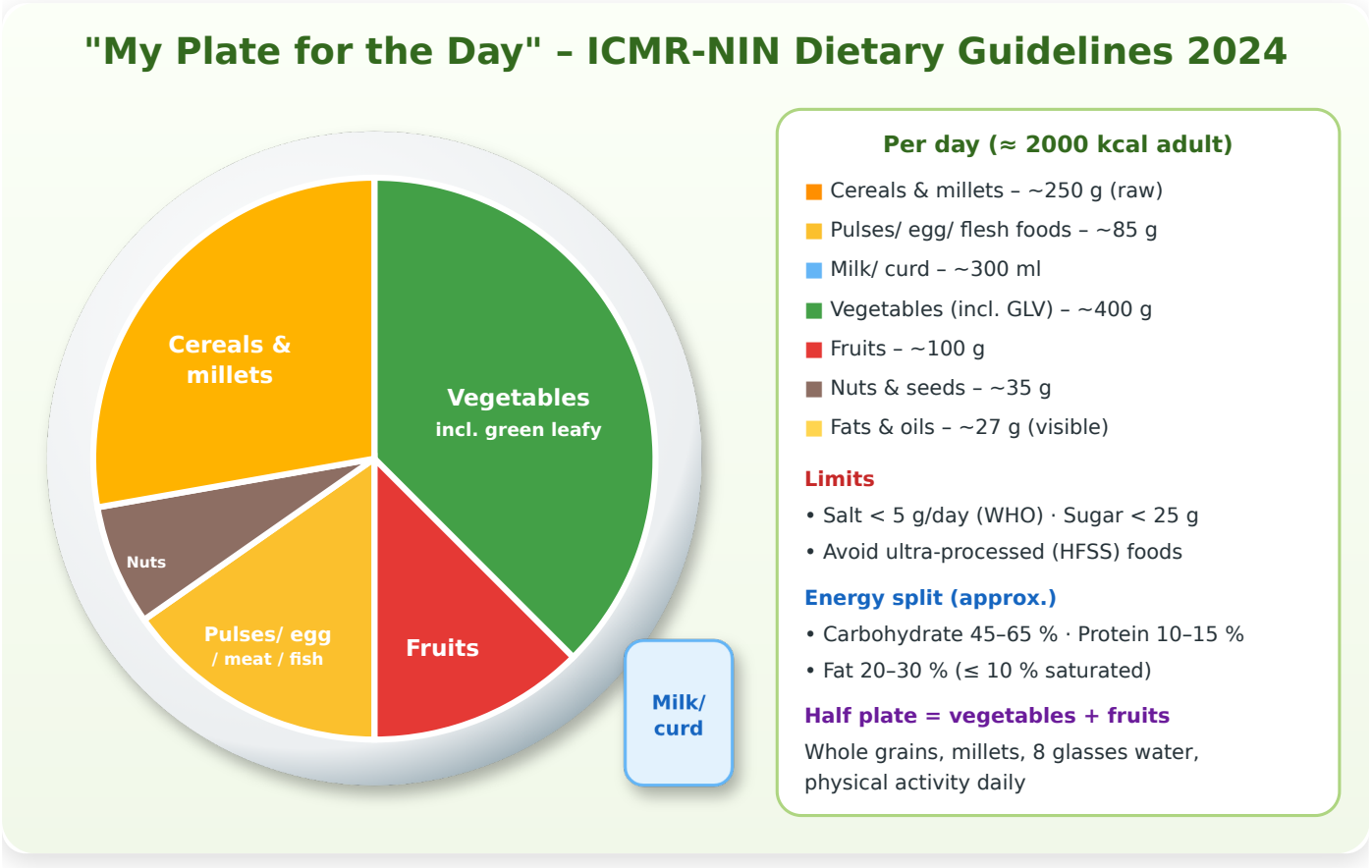
- **EAR (Estimated Average Requirement)** – meets needs of **50 %** of healthy individuals; **RDA (Recommended Dietary Allowance) = EAR + 2 SD – meets needs of ~97.5 %** of population; **TUL (Tolerable Upper Limit)**; **AI**.
- **Reference man (ICMR 2020)** – **age 19–39 yrs, weight 65 kg, height ~177 cm, BMI ~21**, moderately active; **reference woman – 19–39 yrs, 55 kg, ~162 cm** (earlier 2010 – 60 kg & 55 kg).

Group (ICMR 2020)	Energy (kcal/day)	Protein (g/day)
Adult man – sedentary / moderate / heavy	**~2110 / 2710 / 3470**	**~54** (0.83 g/kg)
Adult woman – sedentary / moderate / heavy	**~1660 / 2130 / 2720**	**~46**
Pregnancy (2nd & 3rd trimester)	**+350 kcal**	**+9.5 g (2nd), +22 g (3rd)**
Lactation (0–6 months / 6–12 months)	**+600 / +520 kcal**	**+17 / +13 g**

Group (ICMR 2020)	Energy (kcal/day)	Protein (g/day)
Infant 0–6 / 6–12 months	**~90 / 80 kcal/kg**	~1.4 / 1.2 g/kg
Children 1–3 / 4–6 / 7–9 yrs	**~1110 / 1360 / 1700**	~13 / 16 / 23

- **Protein RDA (adult) ≈ 0.83 g/kg/day** (safe level); (older ICMR – 1 g/kg).
- **Fat** – visible fat **~20–30 g/day** (25–30 g) adult; **total fat ~20–30 % of energy**; **saturated < 10 %**, trans fat < 1 %.
- **Water** – ~**2–3 L/day (8 glasses)**; **1 ml/kcal (adults), 1.5 ml/kcal (infants)**.
- **Salt** – **< 5 g/day (WHO)**; iodised salt.
- **Sugar** – < 5–10 % of energy (**< 25 g/day added sugar**).

1.5 Balanced Diet & Dietary Guidelines



ICMR-NIN "My Plate for the Day" (2024) – proportions & daily quantities

- **ICMR-NIN Dietary Guidelines for Indians (2024) – 17 guidelines**, e.g.: eat a variety of foods (balanced diet), extra food & care in pregnancy/ lactation, **exclusive breastfeeding for 6 months & continue till ≥ 2 yrs**, start **complementary foods at 6 months**, adequate & appropriate diets for children & adolescents, **ensure use of vegetables & fruits (half the plate)**, use oils/ fats in moderation, **adequate protein**, **restrict salt**, **limit sugar & ultra-processed (HFSS) foods**, safe & clean foods, **adequate water**, read food labels, **physical activity**, healthy diet for elderly, avoid protein supplements for muscle building.
- **"My Plate for the Day"** – **~45 % energy from cereals & millets (half as whole grains/ millets)**, **~14–15 % from pulses/ egg/ meat**, **~15 % from nuts, fats & milk**; **vegetables (incl. GLV) & fruits ≈ half the plate by volume**.

- **Food pyramid** – base: cereals (eat most); then fruits & vegetables; milk & pulses (moderate); top: fats, oils, sugar (eat least – sparingly).
- **Millets (shree anna)** – jowar, bajra, ragi, kodo, kutki etc. – **International Year of Millets – 2023** (India's proposal); rich in fibre, calcium (ragi), iron, low GI.
- **Principles of meal planning** – meet nutritional requirements, individual needs (age, sex, activity, physiological state), **economics/ budget**, **food habits, culture & religion**, availability/ season, variety, satiety, **appropriate cooking methods**, family size.

1.6 Cooking & Nutrient Conservation

- **Benefits of cooking** – improves digestibility & palatability, destroys microbes & anti-nutritional factors (trypsin inhibitors in soybean, avidin in raw egg white, goitrogens), adds variety.
- **Methods** – **moist heat** (boiling, steaming, **pressure cooking – best for nutrient retention, fast**), **dry heat** (roasting, baking, grilling), **frying** (fat medium – ↑ calories, loss of heat-labile vitamins, reheated oil – trans fats & toxic polar compounds), **microwave**, sautéing.
- **Nutrient losses** – **vitamin C (most labile – heat, air, alkali, cutting, light), thiamine (heat, alkali, washing rice – polishing)**, folate, riboflavin (light); fat-soluble vitamins more stable.
- **Tips** – **wash vegetables before cutting**, **cut into large pieces just before cooking**, **cook with lid**, minimal water & **use cooking water (soups/ dal)**, **don't add soda (sodium bicarbonate) – destroys thiamine & vitamin C**, **don't wash rice repeatedly/ discard rice water**, **parboiled rice** retains B-vitamins (thiamine migrates into endosperm), avoid reheating; **germination/ sprouting ↑ vitamin C & B, ↓ phytates (better iron absorption)**; **fermentation (idli, dosa, curd) ↑ B-vitamins & digestibility**.
- **Anti-nutritional factors** – **phytates (cereals – ↓ iron, zinc, calcium absorption)**, **oxalates (spinach – ↓ calcium)**, **tannins (tea – ↓ iron)**, **goitrogens (cabbage, cauliflower – ↓ iodine uptake)**, **trypsin inhibitors (soybean – destroyed by heat)**, **avidin (raw egg white – binds biotin)**, **BOAA – neurotoxin in Lathyrus sativus (khesari dal) → lathyrism (spastic paralysis of legs)**, **sanguinarine – argemone oil → epidemic dropsy**, **aflatoxin** (fungal).

♫ Yaad karne ki trick

- Calories – **"4-4-9-7"** – carbohydrate 4, protein 4, fat 9, alcohol 7.
- Cereal-pulse – **"Cereal Lacks Lysine, Pulse lacks Methionine" – Khichdi completes it**.
- SDA highest – **"Protein Produces most heat"**.
- RDA = **EAR + 2 SD (97.5 %)**.

PYQ style

- Science of food & nutrients in relation to health – **nutrition**.
- Father of nutrition – **Lavoisier**.
- NIN located at – **Hyderabad**.
- Most essential nutrient – **water**.
- Energy-yielding nutrients – **carbohydrate, protein, fat**.
- Body-building food – **protein**; protective foods – **vitamins & minerals**.
- 1 g fat gives – **9 kcal**; 1 g carbohydrate/ protein – **4 kcal**; alcohol – **7 kcal**.
- 1 kcal = **4.184 kJ**.
- Major component of energy expenditure – **BMR**.
- BMR highest per kg – **infants**; ↓ in – **hypothyroidism, starvation, old age**.
- BMR rises per °C of fever – **~13 %**.
- Highest SDA – **protein (~30 %)**.
- RDA covers – **97.5 % of population (EAR + 2 SD)**.
- ICMR reference man weight (2020) – **65 kg**; woman – **55 kg**.
- Extra energy in pregnancy – **+350 kcal (2nd & 3rd trimester)**.
- Extra energy in lactation (0–6 months) – **+600 kcal**.
- Protein requirement adult – **0.83 g/kg/day**.
- Salt intake limit – **< 5 g/day**.
- Limiting amino acid in cereals – **lysine**; in pulses – **methionine**.
- Richest plant protein – **soybean**; richest calcium cereal – **ragi**.
- Most heat-labile vitamin – **vitamin C**.
- Best cooking method to conserve nutrients – **pressure cooking/ steaming**.
- Soda in cooking destroys – **thiamine & vitamin C**.
- Lathyrism – **khesari dal (BOAA)**; epidemic dropsy – **argemone oil**.
- Avidin (raw egg white) binds – **biotin**.
- International Year of Millets – **2023**.

2 Macronutrients – Carbohydrates, Proteins, Fats, Fibre & Water

2.1 Carbohydrates

- **Composition** – C, H, O $(CH_2O)n$; **main & cheapest source of energy (~55–65 % of energy in Indian diet)**; **4 kcal/g**.
- **Classification**:
- **Monosaccharides** – **glucose (dextrose – blood sugar)**, **fructose (sweetest sugar – fruits, honey)**, **galactose (milk)**.
- **Disaccharides** – **sucrose (table sugar = glucose + fructose)**, **lactose (milk sugar = glucose + galactose – least sweet)**, **maltose (malt = glucose + glucose)**.
- **Oligosaccharides** – raffinose, stachyose (pulses – flatulence).
- **Polysaccharides** – **starch (cereals, tubers – amylose + amylopectin)**, **glycogen (animal starch – liver & muscle)**, **cellulose, hemicellulose, pectin (dietary fibre)**, dextrins.
- **Functions** – **energy (brain & RBCs depend on glucose – ~130 g/day minimum)**, **protein-sparing action**, **prevents ketosis (anti-ketogenic – fat metabolism needs CHO)**, glycogen storage, flavour, fibre for bowel, synthesis of non-essential amino acids, lactose ↑ calcium absorption.
- **Digestion** – salivary amylase → pancreatic amylase → brush-border disaccharidases (**maltase, sucrase, lactase**) → monosaccharides absorbed in **small intestine (jejunum)**.
- **Deficiency** – ketosis, protein breakdown, weight loss, fatigue; **excess** – obesity, dental caries (sugar), hypertriglyceridaemia, type 2 DM.
- **Lactose intolerance** – lactase deficiency → diarrhoea, bloating after milk → **curd/ fermented milk** tolerated better.
- **Glycaemic index (GI)** – rise in blood glucose after a food vs **glucose (GI = 100)**; **high GI ≥ 70** (glucose, white bread, potato, cornflakes, white rice), **medium 56–69**, **low ≤ 55** (whole grains, millets, pulses, most fruits, milk); low-GI foods preferred in **diabetes**; **glycaemic load** = GI × available CHO/ 100.
- **Sweeteners** – nutritive (sugar, honey, jaggery – iron), **non-nutritive/ artificial (saccharin, aspartame – contraindicated in phenylketonuria, sucralose, stevia – natural)**.

2.2 Dietary Fibre

- **Non-starch polysaccharides + lignin – not digested by human enzymes**; fermented partly by colonic bacteria.
- **Soluble fibre** – **pectin, gums, mucilage, β-glucan** (oats, barley, **fenugreek/ methi seeds**, isabgol, fruits – apple, legumes) → **↓ blood cholesterol & glucose** (slows absorption), forms gel.
- **Insoluble fibre** – **cellulose, hemicellulose, lignin** (wheat bran, whole grains, vegetables) → **↑ stool bulk, prevents constipation**, ↓ transit time.
- **Benefits** – **prevents constipation, haemorrhoids, diverticulosis, colon cancer**, ↓ cholesterol (CHD), glycaemic control (DM), satiety (obesity), gallstones; **requirement ~25–40 g/day (ICMR ~40 g/2000 kcal)**; with adequate fluids.
- **Excess** – flatulence, ↓ mineral absorption (phytates).

- **Low-residue/ low-fibre diet** – **acute diarrhoea, before colonoscopy/ bowel surgery, IBD flare, intestinal obstruction**.

2.3 Proteins

- **Composition** – C, H, O, **N (~16 % – nitrogen × 6.25 = protein)**, S; polymers of **amino acids** (peptide bonds); **4 kcal/g**; **"proteios" – of prime importance (Mulder)**.
- **Functions** – **body building (growth, tissue repair – most important)**, **enzymes, hormones (insulin), antibodies (immunity)**, **haemoglobin, plasma proteins (albumin – oncotic pressure)**, transport (transferrin, lipoproteins), **acid–base buffer**, muscle contraction (actin, myosin), structural (collagen, keratin), energy (when CHO/ fat insufficient), **neurotransmitters**, blood clotting (fibrinogen).
- **Essential amino acids (9) – can't be synthesised – must come from diet**: **histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, valine** – "**PVT TIM HALL**" (P-henylalanine, V-aline, T-hreonine, T-ryptophan, I-soleucine, M-ethionine, H-istidine, A-rginine*, L-eucine, L-ysine) – **arginine & histidine semi-essential (needed in childhood)**.
- **Complete (first-class) proteins** – contain all EAAs in adequate amounts – **animal proteins (egg, milk, meat, fish)** + **soybean**; **incomplete (second-class)** – plant proteins (lack ≥ 1 EAA – **limiting amino acid**).
- **Reference protein – egg (whole egg protein; BV ~94–100, NPU ~96)**; **breast milk** – reference for infants.
- **Protein quality indices** – **biological value (BV – % of absorbed N retained)**, **digestibility coefficient**, **NPU (net protein utilisation = BV × digestibility – best overall)**, **PER (protein efficiency ratio – weight gain per g protein – rats)**, **amino acid (chemical) score**, **PDCAAS/ DIAAS**.
- **Nitrogen balance** – **positive** (growth, pregnancy, convalescence, body building), **negative** (starvation, fever, burns, trauma, surgery, cancer, immobilisation), **equilibrium** (healthy adult).
- **Sources** – **animal** (egg ~13 %, meat ~20 %, fish ~20 %, milk ~3.2 %, paneer ~18 %, cheese ~24 %), **plant** (**soybean ~40–43 % – highest**, pulses ~20–24 %, groundnut ~25 %, cereals ~7–12 %).
- **Deficiency** – **PEM (kwashiorkor, marasmus)**, poor growth, oedema (↓ albumin), poor wound healing, immunodeficiency, anaemia, fatty liver; **excess** – renal load (avoid in CKD), ↑ calcium loss, gout risk (purine-rich).
- **Protein restriction** – **hepatic encephalopathy (acute), CKD (pre-dialysis 0.6–0.8 g/kg)**; **↑ protein** – **burns (1.5–2 g/kg or more), surgery, pregnancy, lactation, nephrotic syndrome (moderate), TB, fever, cancer, dialysis (1.2 g/kg)**.

2.4 Fats (Lipids)

- **Most concentrated source of energy – 9 kcal/g**; esters of **fatty acids + glycerol (triglycerides ~95 % of dietary fat)**.
- **Functions** – energy & **storage (adipose)**, **carrier of fat-soluble vitamins (A, D, E, K)**, **essential fatty acids**, **insulation & protection of organs**, **satiety (slow gastric emptying)**, palatability, **cell membranes (phospholipids)**, synthesis of **steroid hormones, bile acids, vitamin D** (cholesterol), myelin sheath, prostaglandins.

Fatty acids	Features	Sources
Saturated (SFA) – no double bond	**Solid at room temp**; **↑ LDL cholesterol (atherogenic)**	**Ghee, butter, vanaspati, coconut oil (~90 % SFA – highest among oils), palm oil, meat fat, cheese**
Monounsaturated (MUFA) – 1 double bond (**oleic acid – ω-9**)	↓ LDL, maintain HDL	**Olive oil, groundnut oil, mustard oil, rice bran oil, sesame, almonds**
Polyunsaturated (PUFA) – ≥ 2 double bonds	Liquid; **↓ cholesterol**	**ω-6 – linoleic acid (LA)**: sunflower, safflower, corn, soybean oils; **ω-3 – α-linolenic acid (ALA)**: **flaxseed (alsi), mustard, soybean, walnuts, fenugreek**; **EPA & DHA – fatty fish (sea fish), fish oil**
Trans fatty acids	**Partially hydrogenated oils (vanaspati, bakery, fried & reheated oils, fast foods)**; **↑ LDL & ↓ HDL – most harmful**	**FSSAI limit – ≤ 2 % of total fat in oils & foods (since 2022)**; WHO "REPLACE" – eliminate

- **Essential fatty acids (EFA)** – **linoleic acid (ω-6) & α-linolenic acid (ω-3)** – can't be synthesised; **arachidonic acid** semi-essential; **deficiency → phrynoderma (toad skin – follicular hyperkeratosis), dermatitis, poor growth, poor wound healing, hair loss**; **DHA – brain & retina development (breast milk, pregnancy)**; ω-3 – **anti-inflammatory, antithrombotic, ↓ triglycerides, ↓ CHD**.
- **Ideal ω-6 : ω-3 ratio ≈ 5–10 : 1**; **use combination/ rotation of 2 or more vegetable oils** (e.g. groundnut + mustard; rice bran + soybean).
- **Cholesterol** – **only in animal foods (egg yolk – richest common source, organ meats/ brain, prawns, butter, ghee)**; **plant foods – zero cholesterol**; body synthesises (~1 g/day – liver); **desirable serum total cholesterol < 200 mg/dl**, **LDL < 100 ("bad")**, **HDL > 40 (men)/ > 50 (women) ("good" – higher is better)**, **triglycerides < 150 mg/dl**; **dietary cholesterol < 300 mg/day** (older advice).
- **Lipoproteins** – **chylomicrons (dietary TG from gut)**, **VLDL (endogenous TG from liver)**, **LDL (cholesterol to tissues – atherogenic)**, **HDL (reverse cholesterol transport – liver; protective)**.
- **Excess fat** – **obesity, atherosclerosis, CHD, hypertension, type 2 DM, gallstones, certain cancers (breast, colon)**, fatty liver; **deficiency** – EFA deficiency, fat-soluble vitamin deficiency, low energy.
- **Rancidity** – oxidation of fats (off-flavour, toxic); prevented by antioxidants (vitamin E, BHA/ BHT), cool dark storage; **reheated oils – avoid (polar compounds, trans fats); FSSAI – Total Polar Compounds ≤ 25 %**.
- **Fat restriction** – **gall bladder disease, pancreatitis, obesity, CHD, hyperlipidaemia, malabsorption/ steatorrhoea** (use **MCT – medium-chain triglycerides – absorbed directly into portal blood, don't need bile/ lipase**).

2.5 Water & Electrolytes

- **~60 % of adult body weight (~75–80 % in newborn)**; **ICF ⅔, ECF ⅓**; loss of **10 % water – serious, 20 % – fatal**.
- **Functions** – solvent & transport, **temperature regulation (sweating)**, lubrication (joints, eyes), digestion, excretion, cell shape, chemical reactions.
- **Requirement** – **~1 ml/kcal (adults ≈ 2–3 L/day), 1.5 ml/kcal (infants)**; ↑ in **fever (↑ 10–12 % per °C), hot climate, exercise, diarrhoea/ vomiting, lactation, burns, polyuria**; restrict in **renal failure**

(oliguric – previous day's output + 500 ml), heart failure, SIADH, hyponatraemia, cirrhosis with ascites.

- **Sources** – **drinking water & beverages (~60 %)**, **food water (~30 %)**, **metabolic water (~10 %, ~300 ml/day – fat oxidation yields the most)**.
- **Dehydration signs** – thirst, dry mucosa, ↓ skin turgor, sunken eyes/ fontanelle, oliguria, dark urine, tachycardia, hypotension, weight loss; **ORS (WHO low-osmolarity – 245 mOsm/L: Na 75, glucose 75 mmol/L)** + **zinc (10 mg < 6 months, 20 mg > 6 months × 14 days)** in diarrhoea.
- **Sodium** – **main ECF cation**; sources – **common salt (NaCl ~40 % sodium; 5 g salt ≈ 2 g sodium)**, pickles, papad, processed foods, baking soda; **excess → hypertension, oedema**; restrict in **HTN, CHF, renal disease, cirrhosis (ascites), PIH/ oedema, steroid therapy**.
- **Potassium** – **main ICF cation**; sources – **banana, coconut water, citrus/ orange juice, potato, tomato, dry fruits (dates, apricots), legumes, GLV**, tender coconut; **restrict in renal failure/ hyperkalaemia (leaching vegetables – soaking in warm water reduces K^+)**; supplement with diuretics (loop/ thiazide).

♫ Yaad karne ki trick

- Essential amino acids – **"PVT TIM HALL"** – Phenylalanine, Valine, Threonine, Tryptophan, Isoleucine, Methionine, Histidine, Arginine (semi), Leucine, Lysine.
- Sweetest sugar – **"Fructose = Fruit = Fantastic sweet"**; least sweet – **Lactose**.
- Fats – **"SAT is solid, UNSAT is liquid; TRANS is TERRIBLE"**.
- HDL – **"H for Happy/ Healthy – keep it High"**; LDL – **"L for Lousy – keep it Low"**.

PYQ style

- Main & cheapest energy source – **carbohydrates**.
- Sweetest sugar – **fructose**; milk sugar – **lactose**; table sugar – **sucrose**.
- Storage form of CHO in animals – **glycogen**.
- Brain's main fuel – **glucose**.
- CHO prevents – **ketosis (anti-ketogenic) & protein breakdown (protein-sparing)**.
- GI of glucose – **100**; low GI – **≤ 55**.
- Soluble fibre ↓ – **cholesterol & blood sugar**; insoluble fibre prevents – **constipation**.
- Fibre requirement – **~25–40 g/day**.
- Protein contains nitrogen – **16 %** (factor 6.25).
- Essential amino acids – **9**.
- Reference protein – **egg**.
- Best index of protein quality – **NPU**.
- Negative nitrogen balance – **fever, burns, starvation, surgery**.
- Highest protein plant food – **soybean (~40 %)**.
- Most concentrated energy source – **fat (9 kcal/g)**.
- Fat-soluble vitamins – **A, D, E, K**.
- Essential fatty acids – **linoleic & α-linolenic acid**; EFA deficiency – **phrynoderma**.
- Highest saturated fat oil – **coconut oil**.
- ω-3 rich – **fish oil, flaxseed, mustard oil**.
- Most harmful fat – **trans fat (vanaspati)**; FSSAI trans fat limit – **2 %**.
- Cholesterol present only in – **animal foods**; richest – **egg yolk**.
- Good cholesterol – **HDL**; bad – **LDL**.
- Desirable total cholesterol – **< 200 mg/dl**.
- MCT oil used in – **fat malabsorption**.
- Water percentage adult – **60 %**; newborn – **75–80 %**.
- Metabolic water maximum from – **fat oxidation**.
- ORS osmolarity (WHO) – **245 mOsm/L**.
- Potassium-rich – **banana, coconut water, orange juice**.
- Restrict sodium in – **hypertension, CHF, renal disease, ascites**.

3 Vitamins

3.1 General

- **Vitamins** – organic compounds required in **small amounts**, **not synthesised (adequately) in body**, essential for metabolism – act as **coenzymes**; term "**vitamine**" – **Casimir Funk (1912)**; **first vitamin discovered – vitamin A (McCollum)**; **first B vitamin – thiamine (Eijkman – beriberi in chickens)**.
- **Fat-soluble (A, D, E, K)** – absorbed with fat (need bile), **stored (liver, adipose)**, deficiency develops slowly, **toxicity possible (A & D)**, not excreted in urine.
- **Water-soluble (B-complex, C)** – not stored (except **B12 – liver stores for 3–5 yrs**), excess excreted in urine, **daily intake needed**, toxicity rare, easily lost in cooking/ washing.

Vitamins – Sources & Deficiency at a Glance

Fat-soluble (A, D, E, K) – stored, toxic in excess

A — **Retinol – carrot, GLV, mango, papaya, liver, milk**
Night blindness (earliest) → Bitot's spots → xerophthalmia → keratomalacia (blindness)

D — **Calciferol – sunlight (skin), fish oil, egg yolk**
Rickets (children – bow legs, rickety rosary)
Osteomalacia (adults)

E — **Tocopherol – vegetable oils, nuts, seeds**
Antioxidant; haemolytic anaemia in preterm, neuropathy; sterility (animals)

K — **Phylloquinone – GLV, gut bacteria**
Bleeding – haemorrhagic disease of newborn (1 mg IM at birth; 0.5 mg if < 1 kg)

Water-soluble (B, C) – not stored

B1 — **Thiamine – whole grains, pulses**
Beriberi (wet – heart; dry – neuritis), Wernicke

B2 — **Riboflavin – milk, eggs, GLV**
Angular stomatitis, cheilosis, magenta tongue

B3 — **Niacin – groundnut, meat, (tryptophan)**
Pellagra – 3 Ds: dermatitis, diarrhoea, dementia

B6 — **Pyridoxine – cereals, meat**
Neuropathy (INH), convulsions in infants

B9 — **Folic acid – GLV, liver, pulses**
Megaloblastic anaemia, neural tube defects

B12 — **Cobalamin – only animal foods**
Pernicious/ megaloblastic anaemia, neuropathy

C — **Ascorbic acid – amla, citrus, guava**
Scurvy – bleeding gums, poor wound healing

B7 — **Biotin – egg yolk (raw white – avidin)**
Dermatitis, hair loss

Richest sources: Vit C – **amla** · Vit A – **fish liver oil** · Vit D – **sunlight** · B12 – **liver** · Folate – **GLV/ liver**

Vitamins at a glance – sources & deficiency diseases

3.2 Vitamin A (Retinol)

- **Forms** – **retinol (preformed – animal)**, retinal, retinoic acid; **β-carotene (provitamin A – plant; 1 RAE = 1 µg retinol = 12 µg β-carotene)**.
- **Functions** – **vision in dim light (rhodopsin/ visual purple in rods – Wald's cycle)**, **maintenance of epithelial tissues (skin, cornea, conjunctiva, mucosa)**, **immunity ("anti-infective vitamin")**, growth, reproduction, bone.
- **Sources** – **richest – fish liver oils (cod, shark, halibut)**, liver, egg yolk, butter, ghee, milk; **carotene – green leafy vegetables (drumstick/ moringa leaves, spinach, amaranth), yellow-orange fruits & vegetables (carrot, mango, papaya, pumpkin, sweet potato)**, **red palm oil**.
- **RDA (ICMR 2020)** – adult man **~900–1000 µg**, woman **~840 µg** RAE/day; **children ~400–600 µg**.

- **Deficiency – xerophthalmia (WHO classification)**: **XN – night blindness (earliest symptom)**, **X1A – conjunctival xerosis**, **X1B – Bitot's spots (foamy, triangular, greyish-white spots on temporal bulbar conjunctiva – most characteristic sign)**, **X2 – corneal xerosis**, **X3A/ X3B – corneal ulceration/ keratomalacia (< ⅓ or ≥ ⅓ cornea – blindness; emergency)**, **XS – corneal scar**, **XF – xerophthalmic fundus**; also **follicular hyperkeratosis (toad skin)**, ↑ infections (measles severity, diarrhoea, respiratory), growth retardation.
- **Treatment (WHO) – on diagnosis: day 1, day 2 & after 2 weeks (14 days) – age-specific dose** (< 6 months 50,000 IU; 6–12 months 1,00,000 IU; > 1 yr 2,00,000 IU).
- **Prophylaxis (India – National Vitamin A Prophylaxis Programme, 1970)** – **9 doses: 1st – 1 lakh IU at 9 months (with MR-1)**, **2nd – 2 lakh IU at 16–18 months (with DPT booster/ MR-2)**, then **2 lakh IU every 6 months till 5 yrs**; **oral, 1 ml = 1 lakh IU** (spoon provided).
- **Hypervitaminosis A** – acute: **raised ICP (pseudotumour cerebri – bulging fontanelle in infants), headache, vomiting, desquamation**; chronic: liver damage, bone pain, alopecia; **teratogenic (isotretinoin – avoid high doses in pregnancy)**; **carotenaemia** – yellow skin (harmless, sclera spared).

3.3 Vitamin D (Calciferol – "sunshine vitamin")

- **Forms** – **D3 (cholecalciferol – skin & animal)**, **D2 (ergocalciferol – plants/ fungi)**; **activation – liver (25-OH D – storage/ measured form) → kidney (1,25-$(OH)_2$D – calcitriol – active hormone; 1α-hydroxylase stimulated by PTH)**.
- **Synthesis** – **7-dehydrocholesterol in skin + UVB (sunlight)** → D3; **sun exposure ~15–30 min, arms & face, 10 AM–3 PM** (ICMR suggests midday), 2–3 times/week (skin type).
- **Functions** – **↑ intestinal absorption of calcium & phosphorus**, **bone mineralisation**, renal reabsorption of Ca, immunity, muscle function.
- **Sources** – **sunlight (best/ main)**, **fish liver oil (richest dietary)**, fatty fish, egg yolk, liver, butter, **fortified milk & oil (FSSAI +F)**; **breast milk – poor in vitamin D → infant supplement 400 IU/day**.
- **RDA** – **600 IU (15 µg) adults & children; 400 IU infants**; 800 IU elderly; **1 µg = 40 IU**.
- **Deficiency** – **rickets (children)**: **craniotabes (earliest – ping-pong ball skull, < 6 months)**, **frontal bossing, delayed fontanelle closure, delayed dentition**, **rickety rosary (costochondral junctions)**, **Harrison's sulcus**, **pigeon chest**, **widening of wrists (earliest/ best seen – lower end of radius & ulna on X-ray – cupping, fraying, splaying)**, **bow legs (genu varum)/ knock knees (genu valgum)**, double malleoli, pot belly, hypotonia, tetany; **lab – ↑ alkaline phosphatase (earliest biochemical), ↓ phosphate, normal/ ↓ calcium, ↑ PTH, ↓ 25(OH)D (< 12 ng/ml deficiency; < 20 insufficiency)**; **osteomalacia (adults – bone pain, proximal myopathy, waddling gait, Looser's zones)**; osteoporosis contribution.
- **Treatment** – **stoss therapy (single large dose 3–6 lakh IU oral/ IM)** or daily 2000 IU × 3 months + calcium.
- **Hypervitaminosis D** – **hypercalcaemia** (anorexia, vomiting, polyuria, constipation, **renal stones, metastatic calcification**).

3.4 Vitamin E (Tocopherol) & Vitamin K

- **Vitamin E** – **α-tocopherol** – **most important lipid-soluble antioxidant** (protects cell membranes, PUFA, RBCs from oxidation); "**anti-sterility vitamin**" (in rats); **sources – vegetable oils (wheat germ oil – richest, sunflower, safflower), nuts, seeds, GLV**; **RDA ~7.5–10 mg**; **deficiency – haemolytic**

anaemia in premature infants, peripheral neuropathy, ataxia, retinopathy (preterm – ROP prevention), myopathy; fat malabsorption.

- **Vitamin K (phylloquinone – K1 plants; menaquinone – K2 bacteria; menadione – K3 synthetic)** – "**anti-haemorrhagic/ coagulation vitamin**" – **needed for synthesis of clotting factors II, VII, IX, X (γ-carboxylation in liver)** + proteins C & S, osteocalcin; **sources – green leafy vegetables (richest), cabbage, cauliflower, soybean, liver; gut bacteria (large intestine)**; **RDA ~55 µg**.
- **Vitamin K deficiency** – **haemorrhagic disease/ vitamin K deficiency bleeding (VKDB) of newborn** (sterile gut, low placental transfer, **breast milk low in vitamin K**) – classic onset **day 2–7** (bleeding from cord, GI, circumcision, intracranial); **prophylaxis – vitamin K1 1 mg IM (0.5 mg if birth weight < 1000 g) at birth – anterolateral thigh**; also in **obstructive jaundice, malabsorption, prolonged broad-spectrum antibiotics, warfarin**; **↑ PT/ INR**.
- **Warfarin** – vitamin K antagonist → **keep vitamin K (GLV) intake consistent**; **antidote – vitamin K**.

3.5 B-Complex Vitamins

Vitamin	Coenzyme / function	Sources	Deficiency
B1 – Thiamine ("anti-beriberi, anti-neuritic")	**TPP – carbohydrate metabolism (pyruvate dehydrogenase, transketolase)**; RDA **~0.5 mg/1000 kcal (~1.2–1.4 mg)**	**Whole grains, outer layer of cereals (rice bran, polishings), unpolished/ parboiled rice, pulses, nuts, yeast, pork**	**Beriberi** – **wet (cardiac – high-output heart failure, oedema)**, **dry (peripheral neuritis – "burning feet", foot/ wrist drop, calf tenderness, squatting test)**, **infantile beriberi (2–4 months, breastfed by deficient mothers – aphonia/ hoarse cry, cardiac failure)**; **Wernicke's encephalopathy (alcoholics – confusion, ataxia, ophthalmoplegia/ nystagmus) – give thiamine before glucose**; Korsakoff psychosis (memory loss, confabulation); **polished rice eaters**
B2 – Riboflavin	**FMN, FAD – oxidation-reduction (energy metabolism)**; destroyed by **light** (milk in clear bottles)	**Milk & milk products (best), eggs, liver, GLV, pulses, yeast**	**Ariboflavinosis – angular stomatitis/ cheilosis (cracks at corners of mouth), glossitis (magenta tongue), corneal vascularisation, photophobia, seborrhoeic dermatitis (nasolabial, scrotal dermatitis)**
B3 – Niacin (nicotinic acid)	**NAD, NADP**; **synthesised from tryptophan (60 mg tryptophan = 1 mg niacin)**; needs B6	**Groundnut (richest plant), meat, fish, liver, whole grains, pulses**	**Pellagra ("rough skin") – "3 Ds": dermatitis (photosensitive – Casal's necklace around neck, glove & stocking), diarrhoea, dementia (+ 4th D – death)**; **maize/ jowar (sorghum)-eating populations** (maize – niacin bound "niacytin", ↓ tryptophan; **jowar – excess leucine**), **Hartnup disease, carcinoid syndrome, INH therapy**; niacin high dose – flushing (lipid therapy)
B5 – Pantothenic acid	**Coenzyme A**	Widespread ("pantos" = everywhere)	**Burning feet syndrome** (rare)

Vitamin	Coenzyme / function	Sources	Deficiency
B6 – Pyridoxine	**PLP – amino acid metabolism (transamination, decarboxylation)**, neurotransmitters (GABA, serotonin), haem synthesis, tryptophan → niacin	Cereals, pulses, meat, fish, egg, vegetables	**Convulsions in infants**, **peripheral neuropathy (INH therapy – give pyridoxine 10–25 mg)**, **sideroblastic/ microcytic anaemia**, glossitis, stomatitis; used for nausea of pregnancy; **excess – sensory neuropathy**
B7 – Biotin (vitamin H)	Carboxylation reactions	Egg yolk, liver, nuts; gut bacteria	**Rare – raw egg white (avidin binds biotin)** – dermatitis, alopecia, glossitis
B9 – Folic acid (folate; "pteroylglutamic acid")	**THF – one-carbon transfer – DNA synthesis & cell division (RBC formation)**	**Green leafy vegetables ("folium" = leaf – richest), liver, pulses, beans, citrus**; destroyed by heat (cooking losses ~50 %)	**Megaloblastic (macrocytic) anaemia (MCV > 100)**, **neural tube defects (spina bifida, anencephaly) in fetus**, glossitis, diarrhoea; **causes – pregnancy, poor diet, alcohol, drugs (phenytoin, methotrexate, cotrimoxazole), haemolysis**; **RDA – adult ~200–220 µg; pregnancy ~570 µg**; **periconceptional 400 µg (0.4 mg) daily from ≥ 1 month before conception till 12 weeks (5 mg if previous NTD)**; **IFA in pregnancy – 60 mg iron + 500 µg folic acid**
B12 – Cobalamin (cyanocobalamin; "extrinsic factor"; contains cobalt)	Methylcobalamin & adenosylcobalamin – **DNA synthesis, myelin formation**, folate recycling; **absorbed in terminal ileum with intrinsic factor (gastric parietal cells)**	**Only animal foods – liver (richest), meat, fish, egg, milk**; **not in plants** (fermented foods/ algae – traces)	**Pernicious anaemia (autoimmune – no IF)**, **megaloblastic anaemia + neurological signs – subacute combined degeneration of spinal cord (posterior & lateral columns – loss of vibration & position sense, ataxia), peripheral neuropathy, dementia**, glossitis (beefy red tongue), **hyperpigmentation of knuckles**; **causes – strict vegetarians (vegans), gastrectomy, ileal resection/ Crohn's, fish tapeworm (Diphyllobothrium), metformin, PPIs, elderly**; **Schilling test** (historical); **Rx – IM hydroxocobalamin/ cyanocobalamin (lifelong in pernicious anaemia)**; RDA ~**2.2–2.5 µg**; **giving folate alone in B12 deficiency corrects anaemia but worsens neurological damage**

3.6 Vitamin C (Ascorbic Acid – "anti-scorbutic")

- **Functions** – **collagen synthesis (hydroxylation of proline & lysine – wound healing, capillaries, bones, teeth, gums)**, **powerful water-soluble antioxidant**, **↑ non-haem iron absorption (reduces $Fe^{3+} \rightarrow Fe^{2+}$)**, immunity, folate → THF, adrenal steroid & neurotransmitter synthesis, carnitine.
- **Sources** – **amla (Indian gooseberry – richest natural source ~600 mg/100 g; heat-stable relatively)**, **guava (~200 mg)**, citrus (orange, lemon, sweet lime), **drumstick leaves**, GLV, cabbage, tomato, capsicum, **sprouted pulses (germination ↑ vitamin C)**; **milk & cereals are poor sources**; **most easily destroyed vitamin** (heat, air/ oxidation, alkali, copper vessels, cutting & storage).

- **RDA** – **~80 mg (men), ~65 mg (women)** (ICMR 2020); **pregnancy +20, lactation +40 mg**; **infants – 25 mg**; **smokers need more**.
- **Deficiency – scurvy** – **swollen, spongy, bleeding gums (if teeth present)**, **perifollicular haemorrhages & corkscrew hair**, petechiae, ecchymoses, **poor wound healing, anaemia**, joint pain/ haemarthrosis, weakness, **"Sailor's disease" (James Lind 1747 – citrus cured scurvy)**; **infantile scurvy (Barlow's disease, 6–12 months – formula/ boiled milk fed) – subperiosteal haemorrhage, "frog-leg" position, pseudoparalysis, painful limbs, scorbutic rosary (sharp, angular – vs rounded rickety rosary)**; X-ray – **Wimberger's ring, white line of Frankel, ground-glass bone**.
- **Treatment** – 100–300 mg/day + dietary sources; **excess** – oxalate kidney stones, diarrhoea.

3.7 Antioxidants

- **Vitamin E (lipid), vitamin C (aqueous), β-carotene, selenium, zinc, copper, manganese, polyphenols (flavonoids – tea, berries), lycopene (tomato)** – neutralise **free radicals** → ↓ ageing, CHD, cancer, cataract.

Yaad karne ki trick

- Fat-soluble – **"ADEK"** – stored, toxic.
- Pellagra – **"3 Ds + D"** – Dermatitis, Diarrhoea, Dementia, Death.
- Xerophthalmia order – **"N-1A-1B-2-3"** – Night blindness → conjunctival xerosis → Bitot's → corneal xerosis → keratomalacia.
- Vitamin K factors – **"1972"** → 10, 9, 7, 2.
- Wernicke – **"Thiamine before glucose"**.
- Rickety rosary – **round/ smooth (D)**; scorbutic rosary – **sharp (C)**.

PYQ style

- Term vitamine – **Casimir Funk**.
- Fat-soluble vitamins – **A, D, E, K**; stored in – **liver**.
- Water-soluble stored in liver – **B12 (3–5 yrs)**.
- Earliest symptom of vitamin A deficiency – **night blindness**; characteristic sign – **Bitot's spots**.
- Vitamin A prophylaxis first dose – **1 lakh IU at 9 months**; total doses – **9**.
- Richest source of vitamin A – **fish liver oil**; plant – **GLV, carrot (β-carotene)**.
- Hypervitaminosis A in infant – **bulging fontanelle (raised ICP)**.
- Sunshine vitamin – **vitamin D**; active form – **calcitriol (kidney)**.
- Earliest sign of rickets – **craniotabes**; earliest biochemical – **↑ alkaline phosphatase**.
- Rickety rosary – **vitamin D deficiency**.
- Vitamin D for breastfed infants – **400 IU/day**.
- Antioxidant vitamin (fat-soluble) – **vitamin E**; haemolytic anaemia in preterm – **vitamin E deficiency**.
- Vitamin K at birth – **1 mg IM**.
- Haemorrhagic disease of newborn – **vitamin K deficiency**.
- Beriberi – **thiamine (B1)**; wet beriberi – **heart failure**.
- Wernicke's encephalopathy – **thiamine deficiency (alcoholics)**.
- Angular stomatitis, magenta tongue – **riboflavin (B2)**.
- Pellagra – **niacin (B3)**; seen in – **maize/ jowar eaters**.
- Casal's necklace – **pellagra**.
- INH neuropathy – **pyridoxine (B6)**.
- Raw egg white – **avidin → biotin deficiency**.
- Neural tube defects – **folic acid deficiency**; periconceptional dose – **400 µg**.
- B12 only in – **animal foods**; absorbed in – **terminal ileum with intrinsic factor**.
- Subacute combined degeneration – **B12 deficiency**.
- Richest source of vitamin C – **amla**.
- Vitamin C ↑ absorption of – **iron**.
- Bleeding spongy gums – **scurvy (vitamin C)**.
- Infantile scurvy – **Barlow's disease**.
- Most easily destroyed vitamin – **vitamin C**.

4 Minerals & Trace Elements

4.1 Classification

- **Macrominerals (> 100 mg/day)** – **calcium (most abundant mineral in body ~1–1.2 kg, 99 % in bones & teeth)**, phosphorus (2nd), magnesium, sodium, potassium, chloride, sulphur.
- **Trace elements/ microminerals (< 100 mg/day)** – **iron, iodine, zinc, copper, fluoride, selenium, manganese, chromium, cobalt (in B12), molybdenum**.
- **Functions** – structural (bones, teeth), **regulation of body fluids & acid–base**, nerve & muscle function, **enzyme cofactors**, **constituents of hormones (iodine – thyroxine) & haemoglobin (iron)**.

4.2 Calcium & Phosphorus

- **Functions** – **bones & teeth**, **blood clotting (factor IV)**, **muscle contraction (incl. heart)**, **nerve transmission**, cell membrane permeability, enzyme activation, hormone secretion.
- **Sources** – **milk & milk products (best – well absorbed; ~120 mg/100 ml cow's milk)**, **ragi (richest cereal ~344 mg/100 g)**, **sesame (til) seeds (~1450 mg)**, **GLV (amaranth, drumstick leaves, colocasia)**, small fish with bones, curry leaves, jaggery, soybean, tofu, almonds; **lime in paan/ chuna**.
- **RDA (ICMR 2020)** – **adults 1000 mg/day**; **pregnancy & lactation – 1000 mg (+ supplement: 500 mg × 2 calcium + 250 IU vit D – from 14 weeks/ 2nd trimester till 6 months postpartum per MoHFW)**; children 1–3 yrs ~500 mg; adolescents ~1050 mg; elderly ~1200 mg.
- **Absorption ↑ by** – **vitamin D (calcitriol)**, **lactose**, acidic pH, protein, **body need (pregnancy, growth)**, PTH; **↓ by** – **phytates (cereals), oxalates (spinach, amaranth)**, **excess fibre**, excess phosphate/ fat, **caffeine**, steroids, alkaline pH, old age.
- **Regulation** – **PTH ↑ serum Ca, calcitonin ↓, calcitriol ↑**; **normal serum calcium 8.5–10.5 mg/dl**.
- **Deficiency** – **rickets, osteomalacia, osteoporosis (post-menopausal women, elderly – DEXA)**, **tetany (hypocalcaemia – carpopedal spasm, Chvostek & Trousseau signs, laryngospasm)**, dental caries, growth retardation; **excess** – hypercalcaemia, kidney stones, constipation, milk-alkali syndrome.
- **Phosphorus** – **2nd most abundant**; bones (Ca:P ~2:1), **ATP, DNA/ RNA, phospholipids**, buffers; widespread (milk, meat, cereals, pulses); **normal 2.5–4.5 mg/dl**; **restrict in CKD (hyperphosphataemia) – phosphate binders**; **refeeding syndrome → hypophosphataemia**.

4.3 Iron

- **Body iron ~3–4 g (men) / ~2–2.5 g (women)**; **~65–70 % in haemoglobin**, myoglobin (~5–10 %), **storage – ferritin & haemosiderin (liver, spleen, marrow ~25–30 %)**, transport – **transferrin**, enzymes (cytochromes, catalase).
- **Functions** – **O_2 transport (Hb), O_2 storage in muscle (myoglobin)**, cellular respiration (cytochromes), immunity, cognitive development.
- **Forms** – **haem iron (animal – meat, liver, fish, poultry – absorbed ~15–35 %)**; **non-haem iron (plant – ~2–10 % absorption)** – **GLV (amaranth, drumstick, fenugreek), jaggery, bajra, ragi, pulses, dates, dried fruits, garden cress (halim) seeds (very rich), rice flakes (poha)**, **iron cooking vessels**; **milk – poor source of iron**.

- **RDA (ICMR 2020)** – **adult man 19 mg**, **woman 29 mg**, **pregnancy 27 mg (+ supplementation)**, lactation 23 mg, adolescents 32 mg (boys)/ 27 mg (girls), children 1–3 yrs ~8 mg; **body loss ~1 mg/day (men), ~1.5–2 mg/day (menstruating women)**.
- **Absorption** – mainly in **duodenum & upper jejunum**, as **ferrous (Fe^{2+})**; regulated by **hepcidin** (liver); **only ~10 % of dietary iron absorbed**; **↑ by – vitamin C (most important enhancer), haem iron/ meat ("meat factor"), gastric HCl, ferrous form, iron deficiency/ pregnancy (need), fermentation, germination**; **↓ by – tea & coffee (tannins/ polyphenols), phytates (cereals, bran), oxalates, calcium/ milk & calcium supplements, antacids/ PPIs/ achlorhydria, eggs (phosvitin), fibre, zinc excess**.
- **Advice** – **take IFA with/ after meals (fewer GI effects) or between meals with lemon/ amla/ vitamin C**; **avoid tea/ coffee 1 hr before & after**; don't take with calcium tablets (gap 2 hrs).

Iron deficiency anaemia (IDA)

- **Commonest nutritional deficiency & commonest anaemia worldwide/ India**; **NFHS-5 (2019–21) – anaemia in ~57 % women (15–49 yrs), ~52 % pregnant women, ~67 % children (6–59 months), ~25 % men**.
- **Causes** – inadequate intake, **↑ demand (pregnancy, growth, lactation)**, **blood loss (hookworm, menorrhagia, GI bleed, piles, frequent pregnancies)**, malabsorption, malaria.
- **Features** – fatigue, weakness, **pallor (conjunctiva, palms, nail beds, tongue)**, breathlessness, palpitation, **koilonychia (spoon nails)**, angular stomatitis, glossitis, **pica (eating mud/ ice – pagophagia)**, **Plummer-Vinson syndrome (dysphagia – oesophageal web)**, poor concentration/ school performance, ↓ immunity, **in pregnancy – preterm, LBW, PPH tolerance ↓, maternal death**.
- **Lab** – **↓ Hb, microcytic hypochromic RBCs (↓ MCV < 80, ↓ MCH), ↓ serum ferritin (< 15 µg/L – earliest & most specific indicator of depleted stores)**, ↓ serum iron, **↑ TIBC**, ↓ transferrin saturation (< 16 %), ↑ RDW.
- **WHO anaemia cut-offs (Hb g/dl)** – **children 6–59 months < 11**, children 5–11 yrs < 11.5, 12–14 yrs & **non-pregnant women < 12**, **pregnant women < 11 (2nd trimester < 10.5)**, **men < 13**; **severity (pregnancy, India)** – mild 10–10.9, moderate 7–9.9, **severe < 7**; very severe < 4.
- **Treatment** – oral iron (**ferrous sulphate – 200 mg tab = 60 mg elemental iron**) **2 tablets/day (therapeutic)** for 3 months + continue 3 months after Hb normal (replenish stores); **expected rise ~1 g/dl/ 2–3 weeks (~0.3 g/dl per week)**; **reticulocyte response in 5–10 days (earliest sign of response)**; **parenteral iron (IV ferric carboxymaltose – FCM, iron sucrose)** for moderate–severe anaemia in pregnancy/ intolerance; **blood transfusion** – severe (< 7 g/dl near term) or decompensation.
- **Side effects of oral iron** – **black stools (harmless)**, constipation, nausea, epigastric pain, metallic taste; **liquid iron → stains teeth (use straw/ dropper on back of tongue)**.
- **Iron overdose/ poisoning (children)** – **desferrioxamine**.
- **Haemochromatosis** – iron overload.

Anaemia Mukt Bharat (AMB, 2018) – "6×6×6" strategy

Beneficiary	IFA prophylaxis
Children 6–59 months	**IFA syrup 1 ml (20 mg iron + 100 µg folic acid) biweekly (twice a week)** – auto-disable dispenser
Children 5–9 yrs	**IFA pink tablet (45 mg iron + 400 µg FA) weekly**

Beneficiary	IFA prophylaxis
Adolescents 10–19 yrs (in & out of school)	**IFA blue tablet (60 mg iron + 500 µg FA) weekly (WIFS – Weekly Iron & Folic acid Supplementation)**
Women of reproductive age (non-pregnant, non-lactating 20–49 yrs)	**IFA red tablet (60 mg + 500 µg) weekly**
Pregnant women	**IFA red tablet (60 mg iron + 500 µg FA) daily from 4th month (14–16 weeks) for 180 days**, & **lactating mothers – 180 days postpartum**

- **6 interventions** – prophylactic IFA, **deworming (albendazole – National Deworming Day: 10 February & 10 August; 200 mg for 1–2 yrs, 400 mg > 2 yrs)**, **intensified year-round BCC**, **testing & treatment (digital Hb meter)**, **mandatory IFA-fortified foods in public programmes**, **addressing non-nutritional causes (malaria, haemoglobinopathies, fluorosis)**; **6 target groups**; **6 institutional mechanisms**.
- **Target (AMB)** – reduce anaemia by **3 percentage points per year**.
- **Delayed cord clamping (≥ 1–3 min)** at birth ↑ infant iron stores; **iron-folic acid in pregnancy started after first trimester**.

4.4 Iodine

- **Body ~15–20 mg (~70–80 % in thyroid gland)**; **essential for thyroid hormones (T3, T4)** → BMR, growth, **brain development (fetus & infancy)**.
- **Sources** – **iodised salt (main source in India)**, **sea food (sea fish, seaweed – richest)**, food grown in iodine-rich soil; **goitrogens (cabbage, cauliflower, radish, mustard, cassava/ tapioca – thiocyanates)** ↓ iodine uptake.
- **RDA** – **adults 150 µg/day**; **pregnancy & lactation 250 µg**; children 90–120 µg.
- **IDD (iodine deficiency disorders) spectrum** – **fetus: abortion, stillbirth, congenital anomalies, cretinism (neurological – deaf-mutism, squint, spastic diplegia; myxoedematous – hypothyroid, dwarfism)**, ↑ perinatal mortality; **neonate – neonatal hypothyroidism, goitre**; **child/ adolescent – goitre, impaired mental & physical development, ↓ IQ (~13.5 points lower)**; **adult – goitre (simple/ endemic), hypothyroidism, impaired mental function, iodine-induced hyperthyroidism**; **"most common preventable cause of mental retardation/ brain damage in the world"**.
- **Endemic goitre area** – total goitre rate **≥ 5 % in school children (6–12 yrs)**; **sub-Himalayan "goitre belt"**; India – **all states/ UTs have IDD-endemic districts**.
- **Indicators** – **total goitre rate (TGR)**, **median urinary iodine concentration (UIC – 100–199 µg/L adequate; best indicator of recent intake)**, **neonatal TSH (> 5 mIU/L in > 3 %)**, household iodised salt.
- **NIDDCP (1992; earlier NGCP 1962)** – **iodised salt – ≥ 30 ppm at production, ≥ 15 ppm at consumer level**; **ban on non-iodised salt for human consumption (2006)**; **double-fortified salt (DFS – iodine + iron)**; **spot testing kit** (blue colour); store salt in **closed container, away from moisture & sunlight; add after cooking**; **Global IDD Prevention Day – 21 October**; **iodised oil (lipiodol)** in severe endemic areas (older).

4.5 Zinc, Fluoride & Others

Mineral	Functions	Sources	Deficiency / excess
Zinc	**Cofactor of >300 enzymes**, **growth, immunity, wound healing, taste & smell, sexual maturation**, DNA synthesis, insulin storage	**Meat, sea food (oysters – richest), milk, whole grains, pulses, nuts**	**Growth retardation/ stunting, hypogonadism, delayed wound healing, loss of taste (hypogeusia), anorexia, diarrhoea, alopecia, dermatitis**; **acrodermatitis enteropathica (genetic – periorificial & acral dermatitis, alopecia, diarrhoea)**; **zinc in diarrhoea – 20 mg/day × 14 days (10 mg < 6 months)** – ↓ duration & severity, prevents recurrence
Fluoride	**Prevents dental caries (fluorapatite – acid-resistant enamel)**, bone strength	**Drinking water (main source)**, tea, sea fish	**Optimum in water – 0.5–0.8 ppm (mg/L); permissible ≤ 1 ppm (BIS – 1–1.5 ppm max)**; **< 0.5 ppm → dental caries**; **> 1.5 ppm → dental fluorosis (mottled, chalky-white/ brown pitted enamel – children)**; **> 3–10 ppm → skeletal fluorosis (stiff spine, crippling, genu valgum, osteosclerosis)**; **Nalgonda technique (alum + lime + bleaching powder) – defluoridation**; **NPPCF (National Programme for Prevention & Control of Fluorosis, 2008–09)**; **endemic in Rajasthan, Gujarat (north – Mehsana, Banaskantha, Kutch), Andhra, Telangana**
Magnesium	Enzyme cofactor (ATP), neuromuscular function, bones	GLV (chlorophyll), whole grains, nuts	**Tetany (with hypocalcaemia), arrhythmias, seizures** (alcoholics, diarrhoea, diuretics); **$MgSO_4$ – eclampsia**
Selenium	**Antioxidant (glutathione peroxidase)**, thyroid hormone metabolism	Sea food, meat, cereals (soil-dependent)	**Keshan disease (cardiomyopathy – China)**, Kashin-Beck disease; **excess – selenosis (hair & nail loss, garlic breath)**
Copper	Iron metabolism (**ceruloplasmin – ferroxidase**), collagen (lysyl oxidase), melanin, myelin	Liver, shellfish, nuts, legumes	**Anaemia (microcytic), neutropenia, bone changes**; **Menkes disease (kinky hair)**; **excess – Wilson's disease (hepatolenticular degeneration – Kayser-Fleischer ring; penicillamine)**; **Indian childhood cirrhosis (copper vessels/ milk)**
Chromium	↑ **insulin action (glucose tolerance factor)**	Whole grains, meat, brewer's yeast	Impaired glucose tolerance
Manganese, molybdenum, cobalt	Enzymes; **cobalt – part of vitamin B12**	–	Rare
Sodium & potassium	See chapter 2	–	–

♫ Yaad karne ki trick

- Iron absorption – **"C helps, Tea stops"** – vitamin C ↑, tea/ coffee ↓.
- AMB colours – **"Pink for kids (5–9), Blue for teens, Red for women & pregnancy"**.
- Fluoride – **"< 0.5 caries, 0.5–0.8 ideal, > 1.5 fluorosis"**.
- Copper excess – **"Wilson = Wire (copper)"**; Zinc – **"Z for zinc = zest (taste) & zip (wound healing)"**.

✎ PYQ style

- Most abundant mineral in body – **calcium**.
- Best source of calcium – **milk**; richest cereal – **ragi**; richest overall common – **sesame (til)**.
- Calcium RDA adult (ICMR 2020) – **1000 mg/day**.
- Calcium absorption ↑ by – **vitamin D, lactose**; ↓ by – **phytates, oxalates**.
- Tetany – **hypocalcaemia**.
- Iron mostly present in – **haemoglobin**; storage form – **ferritin**; transport – **transferrin**.
- Iron absorbed in – **duodenum (ferrous form)**.
- Enhancer of iron absorption – **vitamin C**; inhibitor – **tea (tannins), phytates, calcium**.
- Milk is poor source of – **iron & vitamin C**.
- Iron RDA adult woman (ICMR 2020) – **29 mg**.
- Commonest nutritional deficiency – **iron deficiency anaemia**.
- Earliest indicator of iron depletion – **↓ serum ferritin**.
- Spoon-shaped nails – **iron deficiency (koilonychia)**.
- Anaemia in pregnancy (WHO) – **Hb < 11 g/dl**; severe – **< 7 g/dl**.
- IFA tablet in pregnancy – **60 mg iron + 500 µg folic acid, 180 days**.
- WIFS for adolescents – **blue tablet weekly**.
- AMB IFA syrup for 6–59 months – **biweekly 1 ml**.
- National Deworming Day – **10 February & 10 August**.
- Black stools with iron – **harmless**.
- Reticulocyte response after iron therapy – **5–10 days**.
- Iodine RDA adult – **150 µg**; pregnancy – **250 µg**.
- Most common preventable cause of mental retardation – **iodine deficiency**.
- Iodised salt at consumer level – **≥ 15 ppm**; production – **≥ 30 ppm**.
- Best indicator of recent iodine intake – **urinary iodine**.
- Goitrogens – **cabbage, cauliflower, tapioca**.
- Zinc in diarrhoea – **20 mg × 14 days**.
- Zinc deficiency – **growth retardation, loss of taste, poor wound healing**.
- Optimum fluoride in water – **0.5–0.8 ppm**; dental fluorosis – **> 1.5 ppm**.
- Defluoridation – **Nalgonda technique**.
- Keshan disease – **selenium deficiency**.
- Wilson's disease – **copper excess (KF ring)**.
- Cobalt is part of – **vitamin B12**.

5 Nutrition Through the Life Span

5.1 Pregnancy

- **Extra requirements (ICMR 2020)** – **energy +350 kcal/day (2nd & 3rd trimester)**; **protein +9.5 g (2nd) & +22 g (3rd trimester)**; **iron 27 mg + IFA supplementation**; **folate 570 µg**; **calcium 1000 mg** (+ supplement); **iodine 250 µg**; vitamin C +20 mg; vitamin A ~900 µg; **DHA/ ω-3**; **water ~2.5–3 L**.
- **Weight gain (IOM, by pre-pregnancy BMI)** – underweight (< 18.5) **12.5–18 kg**; **normal (18.5–24.9) 11.5–16 kg** (**Indian average ~10–12 kg**); overweight 7–11.5 kg; obese 5–9 kg; **~1–2 kg in 1st trimester, then ~0.5 kg/week (~2 kg/month)**; **sudden excess gain (> 1 kg/week) – suspect pre-eclampsia**; inadequate gain → IUGR/ LBW.
- **Supplements (MoHFW)** – **folic acid 400 µg (0.4 mg) from preconception till 12 weeks**; **IFA (60 mg iron + 500 µg FA) daily for 180 days from 14–16 weeks + 180 days postpartum**; **calcium 500 mg BD + vitamin D 250 IU from 14 weeks till 6 months postpartum**; **albendazole 400 mg single dose in 2nd trimester**; **Td**; **iodised salt**.
- **Dietary advice** – **"eat for one, not two" – one extra small meal**; small frequent meals (nausea, heartburn), **↑ fibre & fluids (constipation)**, GLV, milk (2–3 glasses), pulses, eggs/ meat, seasonal fruits; **avoid alcohol, tobacco, excess caffeine (< 200 mg/day), raw/ undercooked meat & eggs (Toxoplasma, Listeria), unpasteurised milk, papaya/ pineapple myths – counsel scientifically**; **food fads & taboos** – counsel (e.g. avoiding nutritious foods).
- **Problems** – **nausea/ vomiting** (dry crackers in morning, small meals, ginger, avoid fatty/ spicy), **heartburn** (small meals, don't lie down after eating, head-end up), **constipation** (fibre, fluids, exercise), **pica**, **anaemia** (IFA, iron-rich foods + vitamin C), **gestational diabetes** (MNT, low GI, avoid sugar; insulin/ metformin if needed), **PIH** (normal salt advised; adequate protein & calcium; excess salt avoided).
- **PM Matru Vandana Yojana (PMMVY, 2017)** – **₹5000** in instalments for the first child; **₹6000 for a second child if girl (PMMVY 2.0)** – compensate wage loss & improve nutrition; **Take-home ration (THR) – ICDS** for pregnant & lactating women.

5.2 Lactation

- **Extra – energy +600 kcal (0–6 months), +520 kcal (6–12 months)**; **protein +17 g / +13 g**; **calcium 1000 mg**; **iodine 250 µg**; vitamin A, C (+40 mg), B12, DHA; **fluids ~3 L/day (drink a glass of water/ milk at each feed)**; galactagogues (traditional – methi, garlic, ajwain, gond laddoo – mostly cultural; **frequent suckling is the best galactagogue**).
- **Avoid** – alcohol, smoking, unnecessary drugs, excess caffeine.
- **Average milk output** – **~500–800 ml/day (~750 ml)** in first 6 months; ~600 ml in 6–12 months.

5.3 Breastfeeding

Composition

Milk	Features
Colostrum (first 2–4 days)	**Thick, yellow (β-carotene), small quantity (~10–40 ml/day; ~2–20 ml per feed)**; **rich in protein (~3× mature milk), IgA (secretory), WBCs, lactoferrin, vitamin A, zinc**; **less fat & lactose**; **laxative – helps passage of meconium (↓ jaundice)**; **"first immunisation"** – must not be discarded (pre-lacteal feeds harmful)
Transitional milk	Day ~5–14
Mature milk	**~70 kcal/100 ml (0.67 kcal/ml)**; protein ~1.1 g/100 ml (**whey : casein ~ 60 : 40/ 70 : 30 – easily digestible**), **fat ~3.5–4 g** (DHA, ARA, **lipase**), **lactose ~7 g (higher than cow's milk)**, Ca 30 mg (well absorbed), **iron low but ~50 % absorbed (lactoferrin)**; **low in vitamin D, vitamin K**
Foremilk	Beginning of feed – **watery, more lactose & protein – quenches thirst**
Hindmilk	End of feed – **rich in fat → satiety & weight gain → empty one breast fully before switching**

Per 100 ml	Breast milk	Cow's milk	Buffalo milk
Energy (kcal)	~65–70	~67	~117
Protein (g)	**~1.1**	**~3.2**	~4.3
Fat (g)	~3.4	~4.1	**~6.5 (highest)**
Lactose (g)	**~7.4 (highest)**	~4.4	~5.0
Calcium (mg)	~28	~120	~210
Iron (mg)	~0.2 (well absorbed)	~0.2	~0.2

Advantages of breastfeeding

- **For baby** – **ideal nutrition, easily digested**, **immunological protection (IgA, lactoferrin, lysozyme, bifidus factor – Lactobacillus bifidus, macrophages)** → **↓ diarrhoea, pneumonia, otitis media, NEC, sepsis**; ↓ allergies, ↓ SIDS; **better IQ & cognitive development (DHA)**; ↓ later obesity, T2DM, hypertension; **sterile, at right temperature, always available, free**; **bonding**; **jaw & teeth development**.
- **For mother** – **↓ PPH (oxytocin → uterine involution)**, **lactational amenorrhoea (LAM – natural contraception: exclusive BF, < 6 months postpartum, amenorrhoeic – ~98 % effective)**, **↓ breast & ovarian cancer**, ↓ T2DM, weight loss, bonding, economical, convenient.
- **Hormones** – **prolactin (milk production/ secretion – "milk-making")** & **oxytocin (milk ejection/ let-down reflex – "milk-ejecting"; inhibited by stress, pain, anxiety, doubt)**; **suckling** stimulates both; **FIL (feedback inhibitor of lactation)** – unemptied breast ↓ production.

Recommendations (WHO/ IYCF India)

- **Early initiation – within 1 hour of birth (incl. caesarean – as soon as mother responsive)**; **skin-to-skin contact**; **no pre-lacteal feeds (honey, ghutti, sugar water, animal milk)**.
- **Exclusive breastfeeding (EBF) for first 6 months** – **no water**, no other food/ liquid **(except ORS, drops/ syrups of vitamins, minerals, medicines)**.

- **Continue breastfeeding till 2 years or beyond** with complementary foods.
- **Demand feeding (8–12 times/24 hrs incl. night)**; **night feeds** maintain prolactin.
- **Signs of good attachment** – **mouth wide open, lower lip turned outward (everted), chin touching breast, more areola visible above than below**; **good positioning** – baby's **head & body in straight line, body close to mother, facing breast (nose opposite nipple), whole body supported** (newborn).
- **Signs of effective suckling** – slow deep sucks with pauses, audible swallowing, cheeks round (not drawn in), baby releases breast satisfied.
- **Adequate milk intake** – **urine ≥ 6 times/24 hrs (wet nappies)** after day 4–5, **weight gain ~500 g/month or more (~20–30 g/day)**, regains birth weight by **10–14 days**, sleeps 2–3 hrs between feeds.
- **Burping** after feeds (upright on shoulder); position on side/ back after feed (sleep supine – SIDS).
- **Breastfeeding positions** – **cradle, cross-cradle (preterm/ small), football/ underarm (twins, after LSCS, large breasts), side-lying (night, after LSCS)**.
- **Expressed breast milk (EBM)** – hand expression (**C-hold, press back towards chest wall, compress & release**) or pump; **storage – room temperature (≤ 25 °C) 6–8 hrs, refrigerator (4 °C) 24 hrs (up to 3–5 days per some guidelines), freezer (−18 °C) 3 months**; **thaw in warm water, don't boil/ microwave**; **feed with cup/ paladai/ spoon (not bottle)**; working mothers.
- **Contraindications (rare)** – **galactosaemia (classic) in baby**, maternal **HIV (in settings where replacement feeding is AFASS; in India – EBF with ART)**, **active untreated TB (separate till non-infectious ~2 weeks of therapy; give EBM; baby INH prophylaxis)**, **anticancer drugs, radioactive isotopes**, illicit drugs, HSV lesion on breast; **NOT contraindicated – maternal hepatitis B (vaccinate baby + HBIG), fever, common cold, mastitis (continue feeding), jaundice in baby (continue)**.

Breast problems

Problem	Management
Sore/ cracked nipples	**Correct positioning & attachment (main cause – poor attachment)**; apply **hindmilk** after feeds, air dry, start on less sore side; don't use soap
Breast engorgement (day 3–5)	**Frequent feeding (don't stop), express milk**, **warm compress before & cold after feeds**, gentle massage, supportive bra, analgesics
Mastitis	**Continue breastfeeding (from both sides)**, empty breast, rest, fluids, antibiotics (cloxacillin), analgesics
Breast abscess	Incision & drainage/ aspiration; feed from other breast, express affected side; resume when healed
Flat/ inverted nipple	**Syringe technique (cut 10/ 20 ml syringe)** after delivery, attachment to areola; antenatal nipple exercises not recommended
Insufficient milk (perceived)	Frequent suckling, night feeds, EBF, relaxation, support (**relactation**)

- **Baby-Friendly Hospital Initiative (BFHI, 1991)** – **"Ten steps to successful breastfeeding"**; **MAA (Mothers' Absolute Affection) programme (2016)**; **World Breastfeeding Week – 1–7 August**; **IMS Act (Infant Milk Substitutes, Feeding Bottles & Infant Foods Act, 1992; amended 2003)** – bans promotion of formula & feeding bottles.

5.4 Complementary Feeding (6–24 months)

- **Start at completed 6 months (180 days)** – breast milk alone insufficient for energy, iron, zinc, vitamin A after 6 months; **continue breastfeeding**.
- **Principles (IYCF – "FATVAH")** – **F**requency, **A**mount, **T**hickness (consistency), **V**ariety (≥ 4–5 food groups – minimum dietary diversity), **A**ctive/ responsive feeding, **H**ygiene.

Age	Texture	Frequency (+ breastfeeding)	Amount per meal
6–8 months	**Thick mashed/ semi-solid (dal-rice, khichdi, mashed banana/ potato, suji kheer)**	**2–3 meals/day** (+ 1–2 snacks)	**Start 2–3 tablespoons → ½ katori (250 ml bowl)**
9–11 months	**Finely chopped/ mashed, finger foods**	**3–4 meals** + 1–2 snacks	**½ katori**
12–24 months	**Family foods (chopped/ mashed)**	**3–4 meals** + 1–2 snacks	**¾ to 1 katori**
Non-breastfed	Add 1–2 extra meals + **1–2 cups milk**	–	–

- **Energy needs from complementary food** – **~200 kcal (6–8 m), ~300 kcal (9–11 m), ~550 kcal (12–23 m)**.
- **Energy-dense** – **add 1 teaspoon of oil/ ghee to each meal**, sugar/ jaggery; **thick (not watery – dal water/ rice water is low in energy)**; **"spoon test" – food should stay on spoon**.
- **Include** – **animal foods (egg, fish, meat), pulses, GLV & yellow fruits/ veg (vitamin A), iron-rich foods**; **introduce one new food at a time (3–5 days)**; **avoid salt & sugar excess, honey < 1 yr (botulism), tea/ coffee, packaged junk foods, whole nuts (choking)**; **cow's milk as main drink not before 1 yr**.
- **Malted/ germinated flours (amylase-rich food – ARF)** – ↓ viscosity → **more energy-dense liquid gruel**; **roasting** (ready-to-use mixes); **fermentation**.
- **Responsive feeding** – feed slowly & patiently, encourage, **feed directly to infants, help older children**; minimise distractions (no TV/ mobile); **separate plate/ bowl** to monitor amount.
- **Hygiene** – wash hands (mother & child), **clean utensils, cup & spoon (not bottle)**, freshly prepared food (**serve within 1–2 hrs**), safe water.
- **Feeding during illness** – **continue breastfeeding more often**, offer soft favourite foods; **after illness – extra meal/day for ≥ 2 weeks (catch-up growth)**.
- **Weaning foods examples (India)** – **khichdi, dalia, suji upma, ragi porridge, mashed banana, sattu, idli, multi-mix (cereal + pulse + oil + jaggery + vegetables) – e.g. "Bal Amrutham"**, THR (ICDS).
- **Bottle feeding hazards** – **diarrhoea (contamination), malnutrition (over-dilution), otitis media, dental caries ("nursing-bottle caries"), nipple confusion, ↓ bonding, expense**.

5.5 Low Birth Weight & Preterm Feeding

- **LBW < 2500 g**; **VLBW < 1500 g**; **ELBW < 1000 g**.
- **Feeding method by gestation/ ability** – **≥ 34 weeks/ good suck – direct breastfeeding**; **32–34 weeks – paladai/ spoon/ cup with EBM**; **28–31 weeks – orogastric/ nasogastric tube (gavage) EBM**; **< 28 weeks/ < 1200 g/ sick – IV fluids then trophic feeds** → advance.

- **EBM (mother's own milk) – best**; **donor human milk (human milk banks – Lactation Management Units/ "Amrit")**; **human milk fortifier** for VLBW; **preterm formula** if no breast milk.
- **Kangaroo Mother Care (KMC)** – skin-to-skin, exclusive breastfeeding, early discharge & follow-up.
- **Supplements for preterm/ LBW** – **vitamin D 400 IU/day**, **iron 2 mg/kg/day from 2–6 weeks (till 1 yr)**, calcium & phosphorus (VLBW), multivitamin.
- **Daily fluid/ feed advancement** – day 1 ~60–80 ml/kg → increase ~20 ml/kg/day up to **150–180 ml/kg/day**.

5.6 Preschool, School-Age & Adolescents

- **Preschool (1–5 yrs)** – **"vulnerable/ most nutritionally at-risk group" (PEM, vitamin A deficiency, anaemia)**; slow growth vs infancy; **food fads, picky eating, poor appetite**; small frequent meals (5–6), family foods, milk, egg, fruits; growth monitoring (MCP card – WHO charts).
- **School-age (6–12 yrs)** – energy for growth & activity; **don't skip breakfast**; healthy tiffin; limit junk/ HFSS foods & sugary drinks; **PM POSHAN (mid-day meal)**; school health – IFA, deworming, vision, dental.
- **Adolescents (10–19 yrs)** – **growth spurt (2nd fastest growth after infancy; ~20 % adult height, ~50 % adult weight gained) → ↑ energy, protein, calcium (bone mass – peak bone mass by ~25–30 yrs), iron (menstruation, ↑ muscle)**; **anaemia** common (girls), eating disorders, junk food, obesity, **early marriage/ pregnancy risks**; **WIFS, RKSK (Rashtriya Kishor Swasthya Karyakram, 2014)**, **Scheme for Adolescent Girls (SAG – ICDS)**.

5.7 Adults & Elderly

- **Adults** – maintain healthy weight (BMI 18.5–22.9 for Asians), balanced diet, ↓ salt/ sugar/ fat, ↑ fibre, physical activity ≥ 150 min/week (moderate).
- **Elderly (≥ 60 yrs)** – **↓ BMR & activity → ↓ energy needs (~10–20 % less)**; **same/ ↑ protein (1–1.2 g/kg – prevent sarcopenia)**, **↑ calcium (1200 mg) & vitamin D (800 IU)**, B12 (↓ absorption – atrophic gastritis), fibre & fluids (constipation – ↓ thirst sensation → dehydration), zinc, antioxidants.
- **Problems** – **poor dentition/ dentures (soft foods)**, ↓ taste & smell, dysphagia, ↓ appetite, **constipation**, loneliness/ depression, poverty, chronic diseases, **polypharmacy (drug–nutrient interactions)**, malabsorption, **osteoporosis**, sarcopenia, dehydration.
- **Advice** – **small frequent soft meals**, nutrient-dense, adequate fluids (6–8 glasses even without thirst), fibre, social eating, **MNA (Mini Nutritional Assessment)** screening.

♫ Yaad karne ki trick

- Breastfeeding – **"1-6-2"** – initiate within **1 hr**, exclusive **6 months**, continue till **2 yrs**.
- Good attachment – **"Wide, Out, Chin, More above"** – mouth wide open, lower lip out, chin touches breast, more areola above.
- Complementary feeding – **"FATVAH"** – Frequency, Amount, Thickness, Variety, Active feeding, Hygiene.
- Foremilk – **thirst**; Hindmilk – **hunger (fat)**.
- EBM storage – **"6-24-3"** – 6 hrs room, 24 hrs fridge, 3 months freezer.

PYQ style

- Extra energy in pregnancy – **+350 kcal**; lactation (0–6 m) – **+600 kcal**.
- Normal weight gain in pregnancy (normal BMI) – **11.5–16 kg** (Indian avg ~10–12 kg).
- IFA in pregnancy – **180 days from 2nd trimester**.
- Folic acid for NTD prevention – **400 µg preconception**.
- Calcium supplement in pregnancy – **500 mg twice daily from 14 weeks**.
- Initiation of breastfeeding – **within 1 hr of birth**.
- Exclusive breastfeeding – **6 months**; continue till – **2 yrs or beyond**.
- Colostrum rich in – **protein, IgA, vitamin A**; called – **first immunisation**.
- Main antibody in breast milk – **secretory IgA**.
- Milk with highest lactose – **human milk**; highest fat – **buffalo milk**; highest protein – **buffalo milk** (human milk lowest ~1.1 g).
- Breast milk low in – **vitamin D, vitamin K**.
- Fat-rich milk at end of feed – **hindmilk**.
- Milk secretion hormone – **prolactin**; ejection – **oxytocin**.
- Let-down reflex inhibited by – **stress, pain, anxiety**.
- Sign of good attachment – **lower lip everted, chin touching breast**.
- Adequate milk – **urine ≥ 6 times/day**.
- Cracked nipple main cause – **poor attachment**.
- Engorgement management – **frequent feeding & expression**.
- Mastitis – **continue breastfeeding**.
- EBM in fridge – **24 hrs**; room temp – **6–8 hrs**.
- EBM fed by – **cup/ paladai, not bottle**.
- Complementary feeding starts – **at 6 months**.
- Add to each complementary meal – **1 tsp oil/ ghee (energy density)**.
- Honey avoided < 1 yr – **infant botulism**.
- Bottle feeding major hazard – **diarrhoea**.
- LBW – **< 2500 g**; 32–34 weeks feeding – **paladai/ spoon**.
- World Breastfeeding Week – **1–7 August**.
- IMS Act – **1992 (amended 2003)**.
- MAA programme – **2016**; BFHI – **1991**.
- Growth spurt nutrition need ↑ – **adolescence**.
- Elderly need extra – **calcium, vitamin D, B12, protein, fluids**.

6 Malnutrition & Nutritional Assessment

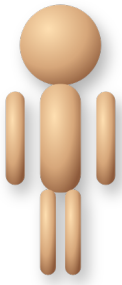
6.1 Forms of Malnutrition

- **Undernutrition** – **wasting** (acute – low weight-for-height), **stunting** (chronic – low height-for-age), **underweight** (low weight-for-age – composite), **micronutrient deficiencies ("hidden hunger" – iron, iodine, vitamin A, zinc)**.
- **Overnutrition** – **overweight & obesity**, diet-related NCDs.
- **"Double/ triple burden of malnutrition"** – coexistence of undernutrition, micronutrient deficiency & overweight in same community/ household.
- **Causes** – **immediate: inadequate dietary intake & disease (infection – vicious cycle "malnutrition–infection")**; **underlying: household food insecurity, inadequate care practices (feeding, hygiene), poor health services & WASH**; **basic: poverty, illiteracy, gender inequality, large family, social & political factors** (**UNICEF conceptual framework**).
- **Ecology** – **"malnutrition–infection cycle"** – infection ↓ appetite, ↑ losses, ↑ requirement → malnutrition → ↓ immunity → infection (measles, diarrhoea, TB, worms).

6.2 Protein-Energy Malnutrition (PEM)

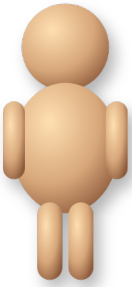
Protein-Energy Malnutrition & SAM Screening

Marasmus



- Severe energy deficit; < 1 yr
- "Skin & bones", old-man face
- NO oedema; alert, hungry
- Wt-for-age < 60 %

Kwashiorkor



- Protein deficit; 1–4 yrs (weaning)
- OEDEMA (pitting – feet), moon face
- Flaky-paint skin, flag sign hair
- Apathy, fatty liver, pot belly

MUAC tape (6–59 months)



Red < 11.5 cm → SAM
Yellow 11.5–12.5 cm → MAM
Green > 12.5 cm → normal

SAM (WHO) – any one:
- Weight-for-height < −3 SD
- MUAC < 11.5 cm
- Bilateral pitting oedema

Complicated SAM → NRC
Uncomplicated → community (CMAM)

WHO 10 steps of SAM management (NRC)

1. Treat/ prevent hypoglycaemia 2. Hypothermia 3. Dehydration (ReSoMal) 4. Electrolytes (K, Mg)
5. Infection (antibiotics) 6. Micronutrients (no iron initially) 7. Cautious feeding (F-75)
8. Catch-up growth (F-100/ RUTF) 9. Sensory stimulation 10. Prepare for follow-up

Indicators: Stunting = low height-for-age (chronic) · Wasting = low weight-for-height (acute) · Underweight = low weight-for-age · Gomez (wt-for-age %) · IAP · Waterlow classification

Marasmus vs kwashiorkor, MUAC tape & SAM criteria, WHO 10 steps

Feature	Marasmus	Kwashiorkor
Cause	**Severe deficiency of energy (& protein) – "starvation"**	**Mainly protein deficiency with relatively adequate calories** ("**the disease of the deposed child**" – **Cicely Williams**, Ga language, Ghana – when next child born, weaning on starchy foods)
Age	**< 1 yr (early weaning, bottle feeding, diarrhoea)**	**1–4 yrs (after weaning)**
Oedema	**Absent**	**Present – pitting, feet & legs → generalised (hallmark)**
Weight	**< 60 % of expected (severe wasting)**	60–80 % (masked by oedema)
Muscle wasting	**Severe – "skin & bones", loose skin folds (baggy pants buttocks)**	Present but masked
Subcutaneous fat	**Absent**	Present (some)
Face	**Old man/ monkey face (shrunken), sunken eyes, prominent cheek bones**	**Moon face (puffy)**
Mental state	**Alert, irritable, hungry (good appetite)**	**Apathetic, miserable, irritable, anorexic**
Hair	Thin, dry	**Sparse, thin, easily pluckable, depigmented (reddish-brown/ pale) – "flag sign" (alternate bands of light & dark – intermittent nutrition)**
Skin	Dry, wrinkled	**"Flaky-paint"/ crazy-pavement dermatosis (hyper/ hypopigmented peeling patches)**
Liver	Not enlarged	**Enlarged – fatty liver**
Serum albumin	Normal/ slightly ↓	**Markedly ↓ (< 2.5 g/dl)**
Prognosis	Better	**Worse (higher mortality)**

- **Marasmic kwashiorkor** – features of both (severe wasting + oedema).
- **Common complications** – **hypoglycaemia, hypothermia (main killers)**, dehydration & electrolyte imbalance (↓ K^+, ↓ Mg^{2+}), **infections (often without fever)**, **anaemia**, **vitamin A deficiency (xerophthalmia)**, heart failure (overhydration, transfusion), **refeeding syndrome**.

Classifications of malnutrition

Classification	Basis	Grades
Gomez (1956)	**Weight-for-age (% of reference median)**	Normal > 90 %; **Grade I 75–90 %; II 60–75 %; III < 60 %**
IAP (Indian Academy of Paediatrics, 1972)	**Weight-for-age (% of standard)**	Normal > 80 %; **Grade I 71–80 %; II 61–70 %; III 51–60 %; IV ≤ 50 %**; add "K" if oedema
Waterlow	**Weight-for-height (wasting) & height-for-age (stunting)**	Wasting: normal > 90 %, mild 80–90, moderate 70–80, severe < 70 %; Stunting: normal > 95 %, mild 90–95, moderate 85–90, severe < 85 %

Classification	Basis	Grades
Wellcome	Weight-for-age + oedema	Kwashiorkor (60–80 % + oedema), marasmus (< 60 %, no oedema), marasmic kwashiorkor (< 60 % + oedema), underweight (60–80 %, no oedema)
WHO (Z-scores/ SD)	**WHO Child Growth Standards (2006 – MGRS)**	**Moderate: −2 to −3 SD; Severe: < −3 SD** for weight-for-height (wasting), height-for-age (stunting), weight-for-age (underweight); **> +2 SD weight-for-height – overweight, > +3 SD – obese**

SAM & MAM (6–59 months)

- **SAM (severe acute malnutrition) – any one**: **weight-for-height/ length < −3 SD (WHO)**, **MUAC < 11.5 cm**, **bilateral pitting oedema** (nutritional origin).
- **MAM (moderate acute malnutrition)** – **WFH between −2 and −3 SD**, or **MUAC 11.5–12.5 cm**.
- **Complicated SAM** (→ **facility-based care – Nutrition Rehabilitation Centre, NRC**) – **failed appetite test** (RUTF), **medical complications** (severe oedema +++, hypothermia, hypoglycaemia, dehydration, high fever, lower respiratory infection, severe anaemia, lethargy/ unconsciousness, convulsions, persistent vomiting), **age < 6 months**; **uncomplicated SAM** (good appetite, no complications) → **community-based management (CMAM)** with **RUTF**/ augmented home food.
- **NRCs** – (since 2008, NRHM/ NHM) – **10–20 beds at district/ CHC**; **average stay ~14 days**; mother stays with child (wage loss compensation, training in feeding & care); follow-up visits (4 visits after discharge – every 15 days); **discharge criteria – 15 % weight gain/ oedema-free for 2 weeks, WFH ≥ −2 SD**.

WHO 10 steps of SAM management

1. **Treat/ prevent hypoglycaemia** (blood glucose < 54 mg/dl – **50 ml of 10 % glucose/ sucrose orally/ NG**; feed 2-hourly day & night).
2. **Treat/ prevent hypothermia** (axillary < 35 °C/ rectal < 35.5 °C – **KMC/ skin-to-skin, cover head, warm room 25–30 °C**; avoid hot water bottles).
3. **Treat/ prevent dehydration** – **ReSoMal (rehydration solution for malnutrition – low sodium 45 mmol/L, high potassium 40 mmol/L)** orally/ NG – **5 ml/kg every 30 min × 2 hrs, then 5–10 ml/kg/hr**; **avoid IV fluids except in shock** (risk of heart failure); signs of dehydration unreliable.
4. **Correct electrolyte imbalance** – **extra potassium & magnesium** (in feeds/ CMV – combined mineral vitamin mix); **↓ sodium** (body sodium is high – don't treat oedema with diuretics).
5. **Treat/ prevent infection** – **broad-spectrum antibiotics routinely** (amoxicillin; parenteral ampicillin + gentamicin if complicated); measles vaccine if unvaccinated (> 6 months).
6. **Correct micronutrient deficiencies** – **vitamin A (day 1 – unless given in last month)**, **folic acid 5 mg day 1 then 1 mg/day**, zinc, copper, multivitamins; **iron NOT in stabilisation phase – start after ~2 weeks (rehabilitation phase, once gaining weight)**.
7. **Start cautious feeding** – **stabilisation phase (day 1–7) – F-75 (75 kcal/100 ml, 0.9 g protein/100 ml) – 130 ml/kg/day (100 ml/kg if severe oedema)**, small frequent feeds (**2-hourly → 3-hourly → 4-hourly**), by cup/ NG.
8. **Achieve catch-up growth** – **rehabilitation/ transition phase – F-100 (100 kcal/100 ml, 2.9 g protein) or RUTF**, gradually ↑ to 150–220 kcal/kg/day; **target weight gain > 10 g/kg/day** (good), 5–10 moderate, < 5 poor.

9. **Provide sensory stimulation & emotional support** – tender loving care, play, structured activities, mother's involvement.
10. **Prepare for follow-up after recovery** – teach mother feeding, hygiene, immunisation, vitamin A, growth monitoring, family planning.
    - **Phases** – **stabilisation (steps 1–7, days 1–7)** → **rehabilitation (weeks 2–6)** → follow-up.
    - **RUTF (ready-to-use therapeutic food)** – **peanut paste + milk powder + oil + sugar + micronutrients (~545 kcal/100 g)**; no water needed, low risk of contamination.
    - **Refeeding syndrome** – too rapid feeding after starvation → **hypophosphataemia (hallmark), hypokalaemia, hypomagnesaemia**, fluid overload, **heart failure, arrhythmias, death** → start slowly (F-75), monitor electrolytes, thiamine.
    - **Appetite test** – offer RUTF (~30–45 min, by mother, quiet place) – **passes if eats ≥ ⅔ of offered/ minimum amount**.

Growth failure indicators

- **Stunting** – chronic undernutrition (long-term, intergenerational, poor sanitation) – **height-for-age < −2 SD**; **NFHS-5: ~35.5 %** under-5s.
- **Wasting** – acute undernutrition (recent weight loss, illness) – **weight-for-height < −2 SD**; **NFHS-5: ~19.3 %**; **severe wasting ~7.7 %**.
- **Underweight** – **weight-for-age < −2 SD**; **NFHS-5: ~32.1 %**.
- **Overweight (children under 5)** – **NFHS-5: ~3.4 %**; adult women overweight/ obese ~24 %.
- **Growth faltering** – earliest sign of malnutrition (flattening of growth curve) – **growth monitoring & promotion (GMP)**.
- **Low birth weight** – reflects maternal nutrition.

6.3 Overweight & Obesity

- **BMI (Quetelet's index) = weight (kg) / height (m^2)**.

BMI	WHO (international)	Asia-Pacific/ Indian (2009 consensus)
Underweight	< 18.5	< 18.5
Normal	18.5–24.9	**18.5–22.9**
Overweight	25–29.9	**23–24.9**
Obese	≥ 30 (I 30–34.9, II 35–39.9, III ≥ 40 – morbid)	**≥ 25**

- **Abdominal (central) obesity** – **waist circumference: Indian men ≥ 90 cm, women ≥ 80 cm** (WHO Europeans – ≥ 102/ 88 cm); **waist-hip ratio > 0.9 (men), > 0.85 (women)**; **"apple shape" (android) – higher CVD/ DM risk** vs "pear shape" (gynoid); **Asian Indian phenotype** – more body fat & central obesity at lower BMI ("thin-fat Indian").
- **Other measures** – **Broca's index (ideal weight = height in cm − 100)**, **Ponderal index**, **Corpulence index (actual/ desirable weight > 1.2 = obese)**, **skinfold thickness (triceps – Harpenden calipers)**, bioelectrical impedance, DEXA.
- **Causes** – positive energy balance (overeating, HFSS foods, sugary drinks, sedentary lifestyle/ screen time), genetics, endocrine (hypothyroidism, Cushing's, PCOS), drugs (steroids, antipsychotics), psychological, ageing.
- **Complications** – **type 2 DM, hypertension, CHD, stroke, dyslipidaemia, metabolic syndrome**, osteoarthritis (knees), **sleep apnoea**, gallstones, fatty liver (NAFLD), GERD, cancers (breast,

endometrial, colon), PCOS/ infertility, varicose veins, psychosocial problems, ↑ surgical & obstetric risk.

- **Management** – **diet: ↓ ~500 kcal/day (weight loss ~0.5 kg/week)**, low-fat, high-fibre, low-GI, adequate protein, ↓ sugar & refined foods, small frequent meals; **physical activity (≥ 150–300 min/week)**; **behaviour modification**; drugs (orlistat, GLP-1 agonists – semaglutide); **bariatric surgery (BMI ≥ 40 or ≥ 35 with comorbidities; Asians lower thresholds)**; **crash diets avoided**.
- **Metabolic syndrome (IDF)** – central obesity + any 2: TG ≥ 150, HDL < 40/50, BP ≥ 130/85, FBS ≥ 100.
- **Eat Right India (FSSAI, 2018)**, **Fit India Movement (2019)**, **"Aaj Se Thoda Kam" (reduce salt, sugar, fat by 10 %)**.

6.4 Nutritional Assessment – "ABCD"

A – Anthropometry

Measurement	Points
Weight	**Most sensitive indicator of current nutritional status/ growth (sensitive to short-term changes)**; **Salter spring balance** (hanging) for children; infant weighing scale; **birth weight doubles at ~5 months, triples at 1 yr, quadruples at 2 yrs**
Height/ length	**Infantometer (recumbent length < 2 yrs)**, **stadiometer (standing > 2 yrs)**; **best indicator of chronic malnutrition (stunting)**; birth length ~50 cm → 75 cm at 1 yr → 100 cm at 4 yrs
MUAC (mid-upper arm circumference)	**Left arm, midpoint between acromion & olecranon**, arm hanging; **"age-independent" between 1–5 yrs (6–59 months – remains ~constant 15–16 cm in well-nourished)**; **Shakir's tape**; **SAM < 11.5 cm, MAM 11.5–12.5 cm, normal > 12.5 cm (13.5 in older)**; **adults (pregnancy) – MUAC < 23 cm = undernutrition**
Head circumference	**At birth ~33–35 cm**, **~45–46 cm at 1 yr**; **> chest circumference till ~6–9 months, equal at 1 yr (~45 cm), then chest > head** (**head/ chest ratio < 1 after 1 yr**; ratio > 1 beyond 1 yr suggests malnutrition)
Chest circumference	**Birth ~31–33 cm**
Skinfold thickness	**Triceps (most common), subscapular, biceps, suprailiac** – **Harpenden/ Lange calipers** – body fat estimate
Waist circumference, WHR, BMI	Adults
Weight-for-height	Wasting (acute); independent of age

- **Growth charts** – **WHO Child Growth Standards (2006)** – used in **Mother & Child Protection (MCP) card**; **growth monitoring: monthly till 1 yr (or 3 yrs ICDS), quarterly till 5 yrs**; **colour bands (green – normal, yellow – moderately underweight (−2 to −3 SD), orange/ red – severely underweight (< −3 SD))**; **"Road to Health" chart (David Morley)**; **direction of growth curve more important than position**; **flat/ falling curve – danger sign**.
- **Velocity** – most children gain **~500–700 g/month (0–6 m), ~250–400 g/month (6–12 m)**.

B – Biochemical tests

- **Hb (anaemia)**, **serum albumin (protein status – long half-life ~20 days; < 3.5 g/dl low)**, **prealbumin/ transthyretin (short half-life 2 days – acute changes)**, transferrin, total lymphocyte

count (immunity), **serum ferritin (iron stores)**, serum retinol (vitamin A), **urinary iodine**, 25(OH)D, B12, folate, **urinary creatinine-height index (muscle mass)**, nitrogen balance, lipid profile, blood glucose/ HbA1c.

C – Clinical examination

- **General** – weight loss, wasting, oedema, apathy; **hair** (flag sign, dull, sparse – PEM), **eyes** (Bitot's, xerosis – vitamin A; pallor – anaemia), **lips** (angular stomatitis, cheilosis – B2), **tongue** (glossitis – B-complex, magenta – B2, beefy red – B12, pale – iron), **gums** (spongy, bleeding – vitamin C), **teeth** (mottling – fluorosis; caries), **skin** (flaky paint – kwashiorkor; phrynoderma – EFA/ vitamin A; Casal's necklace – pellagra; petechiae – vitamin C/ K), **nails** (koilonychia – iron), **thyroid** (goitre – iodine), **bones** (rickety rosary, bow legs – vitamin D), **nervous system** (neuropathy – B1, B12).
- **Jelliffe (1966) – "assessment of nutritional status of the community"** – WHO monograph.

D – Dietary assessment

- **24-hour dietary recall (most common, quick)**, **diet history**, **food frequency questionnaire (FFQ)**, **food diary/ record (3–7 days)**, **weighment method (most accurate – family/ community surveys)**, **inventory method**, **oral questionnaire**; compare with **RDA** – adequacy.
- **Also** – **vital statistics (IMR, under-5 mortality, LBW rate, MMR), ecological factors (food production, socio-economic), health service data**.

Screening tools (hospital)

- **SGA (subjective global assessment)**, **MUST (malnutrition universal screening tool)**, **NRS-2002**, **MNA (elderly)**, **STRONGkids/ STAMP** (children).

Yaad karne ki trick

- Kwashiorkor – **"K for Kidney-like puffiness – oEdema, K for 'Kicked out' child (displaced by sibling)"**; Marasmus – **"M for Muscles gone, Monkey face"**.
- SAM – **"–3, 11.5, Oedema"**.
- Assessment – **"ABCD"** – Anthropometry, Biochemical, Clinical, Dietary.
- Indian BMI – **"23 over, 25 obese"**.
- Waist – **"90 men, 80 women"**.

PYQ style

- Kwashiorkor term given by – **Cicely Williams**; meaning – **disease of the deposed child**.
- Hallmark of kwashiorkor – **oedema**; flag sign – **kwashiorkor**.
- Marasmus – **< 1 yr, no oedema, old-man face, alert & hungry**.
- Fatty liver & low albumin – **kwashiorkor**.
- SAM by MUAC – **< 11.5 cm**; by WFH – **< −3 SD**; bilateral pitting oedema.
- MAM by MUAC – **11.5–12.5 cm**.
- Complicated SAM managed in – **NRC**.
- Main killers in SAM – **hypoglycaemia, hypothermia**.
- ORS for malnourished – **ReSoMal**.
- Iron in SAM – **not in initial (stabilisation) phase**.
- Stabilisation feed – **F-75**; catch-up – **F-100/ RUTF**.
- Good weight gain in rehabilitation – **> 10 g/kg/day**.
- Refeeding syndrome hallmark – **hypophosphataemia**.
- Gomez classification based on – **weight-for-age**.
- IAP grade III – **51–60 %** weight-for-age.
- Stunting – **low height-for-age (chronic)**; wasting – **low weight-for-height (acute)**.
- Most sensitive indicator of nutritional status – **weight**.
- Age-independent measure (1–5 yrs) – **MUAC**.
- MUAC tape – **Shakir's tape**.
- Head = chest circumference at – **1 yr**.
- Birth weight triples at – **1 yr**.
- BMI formula – **weight (kg)/ height (m^2)**; Indian overweight – **23–24.9**; obese – **≥ 25**.
- Central obesity in Indian women – **waist ≥ 80 cm**; men – **≥ 90 cm**.
- Skinfold measured by – **Harpenden calipers (triceps)**.
- Most common dietary survey method – **24-hr recall**; most accurate – **weighment**.
- Protein status marker (short-term) – **prealbumin**.
- Growth chart used in India – **WHO standards (MCP card)**.

7 Therapeutic Nutrition & Diet in Diseases

7.1 Principles of Diet Therapy

- **Therapeutic diet** – modification of normal diet to meet requirements of a disease – in **consistency/ texture, energy, nutrients (protein, fat, CHO, Na, K, fluid), fibre, frequency & timing**.
- **Aims** – maintain/ restore nutritional status, correct deficiencies, rest the affected organ, eliminate harmful foods, adjust to body's ability to use nutrients.
- **Principles** – base on **normal balanced diet**, minimum changes needed, consider **patient's food habits, culture, religion, economic status**, palatability, **small frequent meals**, patient education & follow-up; **dietitian** plans, **nurse** implements & monitors (intake-output, weight, acceptance).

Hospital diets (progressive)

Diet	Foods	Indications
NPO / nil by mouth	Nothing orally	Pre-op (**solids 6–8 hrs, clear liquids 2 hrs**), acute abdomen, GI bleeding, pancreatitis, unconscious
Clear liquid	**Transparent fluids – water, clear soups/ broth, strained fruit juices (no pulp), coconut water, glucose water, tea (no milk), ORS, jelly** – minimal residue, low nutrition (**not > 24–48 hrs**)	**First post-op feed, acute GI upset, before colonoscopy/ bowel prep, severe vomiting/ diarrhoea**
Full liquid	Clear liquids + **milk, milkshakes, custard, curd/ lassi, strained cereal porridge, cream soups, ice cream**	Transition, dysphagia, after mouth/ jaw surgery, inability to chew
Soft diet	**Easily digestible, low fibre, non-spicy, soft texture – khichdi, dalia, idli, mashed potato, boiled egg, cooked vegetables, banana**	**Post-operative transition, fever, GI disorders, chewing difficulty, elderly**
Mechanical soft/ pureed	Minced/ blended foods	Dysphagia (stroke), dental problems
Regular/ normal	Normal balanced diet	No restriction

7.2 Diabetes Mellitus

- **Goals** – **blood glucose control (FBS 80–130, PPBS < 180; HbA1c < 7 %)**, achieve healthy weight, prevent hypoglycaemia & complications, optimal lipids & BP.
- **Energy** – as per ideal body weight & activity (**~20–25 kcal/kg for obese, 30 normal, 35–40 underweight/ active**).
- **Carbohydrate ~45–60 % (complex, low-GI – whole grains, millets, pulses, oats, vegetables)**; **avoid simple sugars (sugar, jaggery, honey, sweets, sweetened drinks, fruit juices)**; **consistent CHO distribution across meals**; **fruits allowed in moderation (whole fruit, not juice; low GI – guava, apple, orange, papaya; limit mango, banana, chikoo, grapes)**.
- **Protein ~15–20 %** (0.8–1 g/kg; restrict in nephropathy); **fat < 30 %** (SFA < 7 %, avoid trans fat); **high fibre 25–40 g/day** (methi seeds, isabgol, oats, salad); **salt < 5 g**.

- **Meal pattern** – **small frequent meals (3 main + 2–3 snacks), regular timing, never skip meals (esp. on insulin/ sulfonylureas)**; **bedtime snack** (insulin – prevent nocturnal hypo).
- **Exchange list system** – foods grouped with similar CHO/ protein/ fat/ energy – allows variety.
- **Hypoglycaemia** – **15–20 g fast-acting CHO (3–4 tsp sugar/ glucose, ½ cup juice, glucose tablets) → recheck in 15 min ("rule of 15")** → then complex CHO snack.
- **Sick-day rules** – continue insulin, take fluids & easily digestible CHO (every 3–4 hrs), monitor glucose & ketones.
- **Gestational DM** – medical nutrition therapy first; avoid sugar, small frequent meals with bedtime snack, adequate calories (don't starve – ketosis).
- **Alcohol** – avoid/ limit (hypoglycaemia); artificial sweeteners in moderation.

7.3 Hypertension & Cardiovascular Disease

- **DASH diet (Dietary Approaches to Stop Hypertension)** – **↑ fruits, vegetables, whole grains, low-fat dairy, nuts, legumes; ↓ saturated fat, red meat, sweets** – rich in **potassium, calcium, magnesium, fibre**; ↓ SBP ~8–14 mmHg.
- **Sodium restriction** – **< 5 g salt/day (~2 g sodium)**; **avoid pickles, papad, chutneys with salt, namkeen/ chips, processed & canned foods, cheese, sauces, baking soda/ powder (bakery items), ajinomoto (MSG), salted butter, dry fish**; **no extra salt at table**; use **lemon, herbs, spices** for flavour; **"low-sodium salt" (KCl) – caution in renal failure**.
- **Mild restriction (no added salt at table) ~2–3 g Na**; **moderate (~1000 mg Na)**; **strict (~500 mg Na – salt-free cooking)** – CHF, ascites.
- **Heart disease (CHD/ dyslipidaemia)** – **low saturated fat (< 7 %), no trans fat, ↓ cholesterol (< 200 mg/day), ↑ fibre (oats, isabgol, methi), ω-3 (fish, flaxseed), plant sterols, nuts**; **mixed oils**; avoid ghee/ butter/ vanaspati/ fried foods, organ meats, egg yolk (limit); **weight reduction**; **limit alcohol, stop smoking**.
- **Myocardial infarction (acute)** – **first 24–48 hrs: liquid/ soft, small frequent feeds, low-sodium, low-fat, easily digestible**, avoid very hot/ cold & caffeine, avoid heavy meals & straining (**stool softeners**).
- **Congestive heart failure** – **sodium restriction (< 2 g Na), fluid restriction (~1–1.5 L/day in severe)**, small frequent meals (prevent abdominal distension), adequate protein, **daily weight (gain > 1 kg/day or 2 kg/week – report)**.

7.4 Renal Disease

Condition	Diet
Acute glomerulonephritis	**Restrict sodium & fluids (oedema, HTN, oliguria)**; **protein restriction if azotaemia**; adequate CHO; **fluid = previous day's urine output + 400–500 ml (insensible loss)**
Nephrotic syndrome	**Normal protein (~1 g/kg; high protein not recommended – worsens proteinuria)**, **sodium restriction (oedema)**, low fat/ cholesterol, adequate energy; no fluid restriction unless severe oedema
Acute kidney injury	Energy 25–35 kcal/kg; protein 0.8–1 g/kg (↑ if dialysis/ catabolic); **restrict K^+, Na^+, phosphate, fluids (output + 500 ml)**

Condition	Diet
CKD (pre-dialysis)	**Low protein (0.6–0.8 g/kg) – mostly high biological value (egg white, milk, fish)**; **adequate energy (30–35 kcal/kg – prevent catabolism)**; **restrict sodium, potassium (avoid banana, coconut water, citrus, tomato, potato, dry fruits; leach vegetables – peel, cut, soak in warm water 2 hrs, boil & discard water), phosphorus (milk, cheese, nuts, colas, whole grains; phosphate binders with meals)**, fluid as per urine output
Haemodialysis	**↑ protein (1.2 g/kg)**, energy 30–35 kcal/kg; restrict K, Na, P, **fluid (interdialytic weight gain < 2–2.5 kg; output + 500 ml)**; water-soluble vitamins (lost in dialysis); **peritoneal dialysis – protein 1.2–1.3 g/kg (protein loss in dialysate), less K restriction**
Renal stones	**↑ fluids (≥ 2.5–3 L/day; urine output ≥ 2 L)** for all; **calcium oxalate – ↓ oxalate (spinach, beetroot, tomato seeds, chocolate, tea, nuts, amaranth), normal dietary calcium (don't restrict), ↓ sodium & animal protein**; **uric acid stones – ↓ purines, alkalinise urine**; **cystine – alkaline urine, fluids**

7.5 Liver & Biliary Disease

- **Acute viral hepatitis** – **high-calorie, high-carbohydrate, moderate protein (1–1.2 g/kg), moderate fat (as tolerated)**; small frequent palatable meals, adequate fluids; **no need for strict fat avoidance/ "no turmeric/ oil" myths**; **avoid alcohol** strictly; sugarcane juice/ glucose for energy.
- **Cirrhosis (compensated)** – **high energy (35–40 kcal/kg), adequate protein (1.2–1.5 g/kg) – protein restriction is NOT routinely recommended**; **late evening snack** (prevent overnight catabolism); **ascites/ oedema – sodium restriction (≤ 2 g Na/ ~5 g salt), fluid restriction only if hyponatraemia (< 125)**; vitamin supplements (B1, folate, fat-soluble); **oesophageal varices – soft foods, avoid rough/ hard/ spicy foods**; zero alcohol.
- **Hepatic encephalopathy** – **avoid prolonged protein restriction**; temporary restriction only in acute severe encephalopathy, then reintroduce; **vegetable & dairy protein preferred (better tolerated)**, **BCAA (branched-chain amino acids)**; **lactulose**; avoid constipation.
- **Gall bladder disease (cholecystitis/ gallstones)** – **low-fat diet (avoid fried, fatty, rich foods, ghee, butter, egg yolk)**; small frequent meals; acute – NPO/ clear liquids; avoid gas-forming foods if bloating.
- **Pancreatitis** – acute: **NPO initially → early enteral feeding (within 24–72 hrs as tolerated) – low-fat**; chronic: **low-fat, small frequent meals, pancreatic enzyme supplements with meals, MCT oil, no alcohol**, fat-soluble vitamins, manage diabetes.

7.6 Gastrointestinal Disorders

Condition	Diet
Peptic ulcer/ gastritis	**Small frequent meals, regular timings**, **avoid foods that cause discomfort – spicy, fried, alcohol, caffeine (tea, coffee, colas), smoking, NSAIDs**; **no late-night meals**; **bland diet not essential (individualised)**; **milk is not a treatment (rebound acid – avoid excessive milk)**
GERD	**Small meals, avoid lying down 2–3 hrs after meals, head-end elevation**, weight loss, avoid fatty foods, chocolate, peppermint, citrus, tomato, coffee, alcohol, carbonated drinks, smoking; **last meal ≥ 3 hrs before bed**

Condition	Diet
Acute diarrhoea	**ORS (low osmolarity) + zinc**; **continue feeding (age-appropriate diet & breastfeeding)** – don't starve; **easily digestible – khichdi, curd rice, banana, boiled potato, dal water, lassi, coconut water**; avoid high-sugar drinks (soft drinks, packaged juices), fatty & spicy foods; **extra meal for 2 weeks after recovery**
Constipation	**High fibre (whole grains, bran, vegetables, fruits with skin, isabgol)**, **fluids 2–3 L**, prunes, figs, exercise, **regular bowel habit (gastrocolic reflex – after breakfast)**
Lactose intolerance	**Avoid/ limit milk; curd, buttermilk, cheese usually tolerated**; lactose-free milk; ensure calcium
Coeliac disease (gluten-sensitive enteropathy)	**Lifelong strict gluten-free diet – avoid wheat, barley, rye (+ oats contaminated) – "BROW"**; **allowed – rice, maize, millets (jowar, bajra, ragi), potato, pulses, sabudana, amaranth (rajgira), buckwheat (kuttu)**
IBD (ulcerative colitis/ Crohn's)	Flare – **low-residue/ low-fibre**, high protein & energy, small meals; remission – normal diet; supplements (iron, B12 – ileal disease, folate, calcium, vitamin D)
Dumping syndrome (post-gastrectomy)	**Small frequent dry meals, avoid simple sugars & fluids with meals (take fluids 30–60 min after/ between meals)**, ↑ protein & fat, **lie down 20–30 min after meals**
Malabsorption/ steatorrhoea	Low fat, **MCT oil**, fat-soluble vitamins, high protein
Irritable bowel syndrome	Low-FODMAP, regular meals, fibre (soluble)
Haemorrhoids/ fissure	High fibre, fluids, sitz bath

7.7 Fever, Infections & Catabolic States

- **Fever** – **↑ energy (BMR ↑ ~13 % per °C)**, **high protein (↑ breakdown – negative N balance)**, **↑ fluids (2.5–3 L; insensible losses ↑)**, electrolytes (Na, K – sweating), vitamins (A, C, B-complex); **soft, easily digestible, small frequent meals** (anorexia); **typhoid – high calorie, high protein, low residue/ fibre (prevent intestinal perforation – avoid roughage), soft diet, plenty of fluids**.
- **Tuberculosis** – **high energy (35–40 kcal/kg), high protein (1.2–1.5 g/kg; ~1.5 g/kg)**, vitamins A, C, D, B6 (INH), calcium, iron, zinc; **Ni-kshay Poshan Yojana (₹1000/month – nutrition support during treatment, since 2024)**; **food baskets – Pradhan Mantri TB Mukt Bharat Abhiyaan (Ni-kshay Mitra)**.
- **HIV/AIDS** – ↑ energy (+10 % asymptomatic, +20–30 % symptomatic), adequate protein, food safety (avoid raw/ undercooked foods), manage diarrhoea/ oral thrush.
- **Burns** – **highest energy & protein needs (hypermetabolism)** – **Curreri formula: 25 kcal/kg + 40 kcal × % TBSA burn**; **protein 1.5–2 (–2.5) g/kg**; **early enteral feeding (within 24 hrs)**; vitamin C, A, zinc (wound healing); fluids (Parkland formula – 4 ml × kg × %TBSA in 24 hrs).
- **Surgery** – **pre-op – correct malnutrition, carbohydrate loading**; **post-op – early oral/ enteral feeding (ERAS – within 24 hrs if possible; bowel sounds not mandatory)**, progress clear → full liquid → soft → normal; **high protein, vitamin C (collagen), zinc (wound healing), vitamin K, iron**.
- **Cancer** – high energy & protein, small frequent meals, manage nausea (dry foods, cold foods, ginger), mucositis (soft, bland, cool, non-acidic), taste changes (plastic utensils for metallic taste), **neutropenic diet (well-cooked foods, avoid raw/ unwashed salads, unpasteurised products)**; cachexia.
- **Pressure ulcers/ wounds** – ↑ protein, energy, vitamin C, zinc, arginine, fluids.

7.8 Other Conditions

Condition	Diet
Gout/ hyperuricaemia	**Low purine – avoid organ meats (liver, kidney, brain), sardines, anchovies, shellfish, red meat, meat extracts/ gravies, yeast, alcohol (beer)**; moderate pulses/ spinach/ mushrooms; **↑ fluids (≥ 3 L)**; low-fat dairy, cherries; avoid fasting & fructose drinks; weight loss gradually
Iron deficiency anaemia	Iron-rich foods (GLV, jaggery, meat, liver, dates, garden cress seeds, ragi, bajra) **+ vitamin C**; avoid tea/ coffee with meals; iron vessels
Osteoporosis	Calcium (1000–1200 mg), vitamin D, protein, weight-bearing exercise, ↓ caffeine/ salt/ alcohol
Phenylketonuria (PKU)	**Low-phenylalanine diet (lifelong, esp. till brain maturity)** – avoid high-protein foods & **aspartame**; special formula (Lofenalac); **breastfeeding allowed partially with monitoring**; tyrosine supplement
Galactosaemia	**Lactose-/ galactose-free diet – stop breastfeeding & milk**; soy-based formula
Maple syrup urine disease	Restrict BCAAs (leucine, isoleucine, valine)
Epilepsy (refractory)	**Ketogenic diet** – high fat (4:1 fat : protein + CHO)
Wilson's disease	Low copper – avoid liver, shellfish, nuts, chocolate, mushrooms
Obesity	Low calorie (−500 kcal/day), high fibre, low fat & sugar
Hypothyroidism / goitre	Iodised salt, avoid excess goitrogens (raw cabbage/ cauliflower)
Pre-eclampsia	Balanced diet, adequate protein & calcium; **normal salt (no strict restriction)**

7.9 Enteral & Parenteral Nutrition

Enteral (tube) feeding – "if the gut works, use it"

- **Preferred over parenteral** – maintains gut integrity (prevents mucosal atrophy & bacterial translocation), fewer infections, cheaper, physiological.
- **Indications** – functional GI tract but unable to eat adequately – **unconsciousness, stroke/ dysphagia, head & neck cancer/ surgery, oesophageal obstruction, burns, critical illness, preterm infants, anorexia**.
- **Routes** – **nasogastric (NG – short-term < 4–6 weeks)**, **nasoduodenal/ nasojejunal (aspiration risk, gastroparesis)**, **gastrostomy (PEG – long-term > 4–6 weeks)**, jejunostomy; **orogastric** in neonates.
- **NG tube length – NEX (nose → earlobe → xiphisternum) (or nose-ear-mid-xiphoid-umbilicus in neonates)**; **confirm position before every feed – X-ray (gold standard, initial), pH of aspirate (≤ 5.5 gastric)**; **auscultation (whoosh test) – unreliable, not recommended alone**.
- **Methods** – **bolus (200–400 ml over 10–15 min, 4–6 times/day by gravity/ syringe)**, **intermittent**, **continuous (pump – critically ill, jejunal feeds)**.
- **Feeds** – **blenderised kitchen feeds (hospital – milk, egg, dal water, cereals, sugar, oil – strained)**, **commercial polymeric formulas (1 kcal/ml)**, elemental/ peptide-based (malabsorption), disease-specific (renal, diabetic).

- **Procedure** – **hand hygiene; patient in Fowler's/ semi-Fowler's position (≥ 30–45°) during & 30–60 min after feed (aspiration prevention)**; check placement & **gastric residual volume** (hold if large – e.g. > 250–500 ml as per policy); feed at **room temperature**, by gravity slowly; **flush with 30 ml water before & after feeds and medications**; never mix drugs with feed; **crush only crushable tablets** (not enteric-coated/ sustained-release); **change feed bag/ set every 24 hrs**; prepared feeds – use within 4 hrs (open system) / refrigerate ≤ 24 hrs; **oral care**; record intake-output, weight.
- **Complications** – **aspiration pneumonia (most serious)**, **diarrhoea (commonest – fast rate, hyperosmolar feed, contamination, antibiotics, cold feed)**, constipation, nausea/ vomiting, tube displacement/ blockage (flush; warm water), nasal erosion, sinusitis, **refeeding syndrome**, hyperglycaemia, electrolyte imbalance, dehydration.

Parenteral nutrition (PN/ TPN)

- **Indications** – **non-functional/ inaccessible GI tract** – **short bowel syndrome, intestinal obstruction/ ileus, high-output fistula, severe pancreatitis (if enteral not tolerated), severe malabsorption, prolonged NPO > 5–7 days**, chemo-induced severe mucositis, preterm (VLBW) infants.
- **Routes** – **central (TPN – hypertonic, > 900 mOsm/L – via CVC/ PICC – subclavian)**; **peripheral (PPN – < 900 mOsm/L, short-term)**.
- **Components** – **dextrose (CHO), amino acids (protein), lipid emulsion (fat – 10/ 20 % – Intralipid; white)**, electrolytes, vitamins, trace elements, water; **3-in-1 (all-in-one) bags**.
- **Nursing care** – **strict asepsis (dedicated lumen – no other drugs/ blood through TPN line)**, **infusion pump, constant rate – never speed up to "catch up"**, **change bag/ tubing every 24 hrs (lipid tubing every 12–24 hrs; lipids alone hang ≤ 12 hrs)**, check bag before hanging (cracking/ oil droplets in 3-in-1 → discard), **protect lipids/ vitamins from light**, store in fridge & warm to room temp before use (remove 1 hr prior), **monitor blood glucose (6-hourly initially), electrolytes, LFT, triglycerides, weight daily, intake-output**, signs of infection; **don't stop abruptly (rebound hypoglycaemia) – taper or hang 10 % dextrose**.
- **Complications** – **catheter-related bloodstream infection (sepsis – commonest serious)**, **hyperglycaemia (commonest metabolic)**, hypoglycaemia (abrupt stop), **refeeding syndrome**, electrolyte imbalance, fluid overload, **air embolism (Valsalva/ Trendelenburg left-side during line change)**, pneumothorax (insertion), thrombosis, **liver dysfunction (cholestasis, fatty liver – PNALD)**, gallstones, gut atrophy.

7.10 Drug–Food Interactions (Selected)

Drug	Interaction / advice
Warfarin	Keep **vitamin K (GLV)** intake consistent
MAO inhibitors	**Avoid tyramine (aged cheese, wine, pickled/ smoked foods, soy sauce)** – hypertensive crisis
Tetracyclines, fluoroquinolones	Avoid with **milk, antacids, iron, calcium** (chelation)
Iron	Avoid tea/ coffee/ milk; take with vitamin C
Levothyroxine	Empty stomach; avoid soy, calcium, iron within 4 hrs
Grapefruit juice	↑ levels of statins, CCBs, cyclosporine

Drug	Interaction / advice
Isoniazid	Pyridoxine supplement; avoid tyramine/ histamine-rich fish
Levodopa	Avoid high-protein meals with dose; avoid pyridoxine supplements
Metformin	Take with meals; long-term **B12 deficiency**
Diuretics (loop/ thiazide)	**K^+-rich foods**; K-sparing – avoid excess K
Alcohol + metronidazole	Disulfiram-like reaction
Steroids	↑ protein, calcium, vitamin D; ↓ sodium & sugar
Lithium	**Maintain consistent salt & fluid intake** (low salt → toxicity)

7.11 Nurse's Role in Nutrition

- **Nutritional screening & assessment** on admission (weight, height, BMI, MUAC, diet history, appetite).
- **Plan with dietitian**, explain diet to patient/ family, respect preferences.
- **Assist feeding** – position (upright), mouth care, prepare environment, assist dependent patients, adequate time; **feeding dysphagic patients – thickened fluids, chin tuck, small bites**.
- **Monitor intake-output, food charts, weight, lab values**; identify & report poor intake, nausea, vomiting.
- **Tube feeding & TPN care**; prevent complications.
- **Health education** – balanced diet, special diets, food hygiene, breastfeeding & complementary feeding, discharge diet counselling.
- **Protected mealtimes**; "red tray" system for patients needing help.

♫ Yaad karne ki trick

- Gluten – **"BROW"** – Barley, Rye, Oats, Wheat.
- Renal diet – **"Low P-K-Na, controlled Protein & Fluids"**.
- Enteral first – **"If the gut works, use it"**.
- Tube feed position – **"Head up 30–45° – Aspiration prevented"**.
- TPN – **"Never catch up, never stop abruptly"**.
- Hypoglycaemia – **"Rule of 15"**.

PYQ style

- First feed after surgery – **clear liquids**.
- Clear liquid diet includes – **water, clear soup, strained juice, coconut water (no milk)**.
- Diabetic diet – **complex CHO, high fibre, low GI, small frequent meals**.
- Hypoglycaemia conscious – **15 g sugar, recheck after 15 min**.
- DASH diet for – **hypertension**.
- Salt limit in HTN – **< 5 g/day**.
- Foods high in sodium – **pickles, papad, processed foods, baking soda**.
- CHF – **sodium & fluid restriction, daily weight**.
- CKD diet – **low protein (HBV), low K, Na, P**.
- Haemodialysis protein – **↑ (1.2 g/kg)**.
- Potassium-rich foods avoided in CKD – **banana, coconut water, orange juice, potato**.
- Leaching reduces – **potassium in vegetables**.
- Nephrotic syndrome – **normal protein, low salt**.
- Renal stones – **↑ fluids ≥ 3 L**; oxalate-rich – **spinach, tomato, chocolate, tea**.
- Hepatitis diet – **high calorie, high CHO, moderate protein & fat**.
- Cirrhosis with ascites – **sodium restriction**.
- Gall bladder disease – **low-fat diet**.
- Peptic ulcer – **small frequent meals, avoid spicy/ caffeine/ alcohol**.
- GERD – **avoid lying down 2–3 hrs after meals**.
- Diarrhoea – **continue feeding + ORS + zinc**.
- Constipation – **high fibre & fluids**.
- Coeliac disease – **gluten-free (avoid wheat, barley, rye, oats)**.
- Dumping syndrome – **dry meals, fluids between meals, lie down after**.
- Typhoid diet – **high calorie, high protein, low residue**.
- Burns – **highest calorie & protein (Curreri formula)**.
- Gout – **low purine (avoid organ meats, red meat, alcohol)**.
- PKU – **low phenylalanine; avoid aspartame**.
- Galactosaemia – **stop milk/ breastfeeding**.
- Refractory epilepsy – **ketogenic diet**.
- Preferred nutrition route – **enteral**.
- NG tube length – **NEX**; confirm placement – **X-ray/ pH ≤ 5.5**.
- Tube feeding position – **Fowler's ≥ 30–45°**; flush with – **30 ml water**.
- Commonest tube-feeding complication – **diarrhoea**; most serious – **aspiration**.
- TPN route – **central vein**; commonest serious complication – **infection**; metabolic – **hyperglycaemia**.
- Lipid emulsion appearance – **milky white**.
- MAOI + cheese – **hypertensive crisis**.
- Warfarin – **consistent vitamin K intake**.

8 Food Hygiene, Safety, Preservation & Adulteration

8.1 Food Hygiene

- **Food hygiene** – all conditions & measures necessary to ensure **safety & wholesomeness of food** at all stages – production, processing, storage, distribution, preparation & consumption.
- **WHO "Five Keys to Safer Food"** – **1. Keep clean** (hands, surfaces, utensils), **2. Separate raw & cooked** (avoid cross-contamination), **3. Cook thoroughly** (core temperature **≥ 70–75 °C**), **4. Keep food at safe temperatures** (**danger zone 5–60 °C** – bacteria multiply fastest; don't keep cooked food at room temp > 2 hrs; refrigerate < 5 °C; reheat to > 70 °C/ steaming hot), **5. Use safe water & raw materials**.
- **Food handlers** – **medical examination & stool test (carriers – typhoid, amoebiasis, Shigella)**; **exclude if diarrhoea, vomiting, fever, jaundice, infected skin lesions/ boils (Staph), sore throat, discharging ears**; **hand washing** (after toilet, before handling food), short clean nails, clean apron & head cover, no smoking/ spitting; health education; **typhoid vaccination**.
- **Kitchen hygiene** – clean premises, screened windows (flies), fly-proof & rodent-proof storage, safe water, proper drainage & waste disposal (covered bins), separate chopping boards (raw meat vs vegetables), clean utensils (wash with detergent & hot water, air-dry).
- **Meat hygiene** – **ante-mortem & post-mortem inspection** at licensed slaughterhouses; diseases transmitted – **taeniasis (T. solium – pork, T. saginata – beef), trichinosis, toxoplasmosis, anthrax, brucellosis, TB (bovine), salmonellosis**; **thorough cooking**; **"measly pork/ beef"** – cysticerci.
- **Fish & egg** – fresh fish – bright eyes, red gills, firm flesh; **egg freshness – sinks in water (fresh), floats (stale)**; candling.

8.2 Milk Hygiene

- **Milk-borne diseases** – **from animals (zoonoses): bovine TB (M. bovis), brucellosis, Q fever, Salmonella, anthrax, FMD**; **from humans/ handlers: typhoid, paratyphoid, cholera, dysentery, hepatitis A, diphtheria, streptococcal infections, Staph food poisoning**.
- **Pasteurisation** – kills pathogens (not all bacteria/ spores):
- **Holder (vat) method – 63–66 °C for 30 min**, then rapid cooling to < 5 °C.
- **HTST (flash) – 72 °C for 15–20 seconds** (most common commercially), cooled to < 4 °C.
- **UHT (ultra-high temperature) – 125–140 °C for 2–4 seconds** – Tetra Pak, long shelf life without refrigeration.
- **Tests** – **phosphatase test (efficiency of pasteurisation – alkaline phosphatase should be destroyed)**, **methylene blue reduction test (bacterial load/ keeping quality)**, standard plate count, coliform count, turbidity test (sterilised milk).
- **Boiling** – simple home method (kills most pathogens).
- **Toned milk (3 % fat)**, **double-toned (1.5 %)**, **standardised (4.5 %)**, **full cream (6 %)**, **skimmed (≤ 0.5 %)** – fat content; **Operation Flood (1970) – Dr Verghese Kurien ("Father of the White Revolution", Amul – Anand, Gujarat)**; **National Milk Day – 26 November**.

8.3 Food-borne Diseases & Food Poisoning

- **Food poisoning** – acute gastroenteritis from food contaminated with **bacteria/ toxins/ chemicals**; **characteristics – sudden outbreak, many people who ate common food, short incubation, similar symptoms**.

Type	Organism / incubation	Key features
Staphylococcal	**S. aureus – preformed enterotoxin (heat-stable)**; **1–6 hrs (shortest)**	**Sudden nausea, severe vomiting, cramps**; no fever; **milk products, sweets, custard, cream, cold meats** handled by **carriers with boils/ nasal**; recovery in 24 hrs
Bacillus cereus	Emetic (preformed – 1–6 hrs) / diarrhoeal (8–16 hrs)	**Fried/ reheated rice ("Chinese restaurant syndrome")**
Salmonella	**Infection – 12–24 hrs (6–72)**	**Fever, diarrhoea, vomiting, abdominal pain**; **poultry, eggs, meat, milk**
Clostridium perfringens	8–24 hrs	Diarrhoea, cramps; **meat stews/ gravies cooled slowly (reheated)**
Clostridium botulinum	**12–36 hrs (2 hrs–8 days)**	**Botulism – neurotoxin – descending flaccid paralysis, diplopia, ptosis, dysphagia, dysphonia, respiratory paralysis; NO fever/ diarrhoea usually**; **home-canned/ preserved low-acid foods, sausages, smoked fish; honey (infant)**; **highest mortality**; **antitoxin**; toxin destroyed by boiling 10 min
Vibrio parahaemolyticus	12–24 hrs	Seafood
E. coli (ETEC, EHEC)	1–3 days	Traveller's diarrhoea; HUS (EHEC – undercooked beef)
Shigella, Campylobacter, Listeria, Norovirus, hepatitis A & E, Cholera	Variable	Water & food

- **Investigation of outbreak** – list of persons who ate & fell ill, **food histories (attack rate – food-specific)**, collect **samples of suspected food, vomitus & stool** for lab, inspect kitchen & handlers, **notify** authorities; **prevention – food hygiene, refrigeration, thorough cooking, health education**.
- **Other food toxins** – **lathyrism (Lathyrus sativus/ khesari dal – BOAA neurotoxin – spastic paralysis of lower limbs; prevention – steeping (soaking) in hot water, parboiling, vitamin C; ban on sale)**; **epidemic dropsy (argemone oil adulteration of mustard oil – sanguinarine → oedema of legs, glaucoma, cardiac failure; nitric acid test – orange-red; paper chromatography)**; **aflatoxicosis (Aspergillus flavus on groundnuts/ maize – liver damage, HCC)**; **ergotism (Claviceps purpurea – rye/ bajra – gangrene, convulsions)**; **endemic ascites (pyrrolizidine alkaloids – Crotalaria seeds in millet)**; **fluorosis (water)**; **mushroom poisoning**, **cassava (cyanide)**, **solanine (green potato)**; **heavy metals – lead, mercury (Minamata), cadmium (itai-itai)**; pesticides.

8.4 Food Preservation

- **Aim** – prevent spoilage by **microbes, enzymes, oxidation, insects**; extend shelf life while retaining nutrients.

Method	Examples
Low temperature	**Refrigeration (0–5 °C; slows growth)**, **freezing (−18 °C or below; deep freezing/ quick freezing)**, cold storage
High temperature	**Pasteurisation**, **sterilisation/ canning (heat + airtight)**, boiling, blanching (enzyme inactivation of vegetables before freezing)
Removal of water (drying/ dehydration)	**Sun drying (papad, grains, fish, chillies), spray drying (milk powder), freeze-drying (lyophilisation – instant coffee)**
Increasing osmotic pressure	**Salting (pickles, salted fish)**, **sugar (jams, jellies, murabba, squash – 65–70 % sugar)**
Chemicals (class II preservatives)	**Sodium benzoate (fruit juices, squashes – acidic foods)**, **potassium metabisulphite (KMS – light-coloured fruits, juices)**, sorbic acid, vinegar (acetic acid), nitrites (meat curing)
Class I preservatives (natural)	**Salt, sugar, vinegar, spices, oil, honey, alcohol, wood smoke**
Fermentation	**Curd, idli, dosa, dhokla, vinegar, sauerkraut** – lactic acid
Smoking	Fish, meat
Irradiation (gamma rays – cold sterilisation)	Onions, potatoes (sprout inhibition), spices, grains; **Radura symbol**
Modern	Vacuum packing, modified-atmosphere packing, aseptic packaging (Tetra Pak)

8.5 Food Additives & Fortification

- **Food additives** – substances added intentionally (in small quantities) to improve **appearance, flavour, texture, shelf life** – **preservatives, colours (permitted synthetic – tartrazine, sunset yellow; not in sweets beyond limit; natural – turmeric, saffron, caramel)**, **flavours, sweeteners (aspartame – PKU warning)**, **antioxidants (BHA, BHT – in oils)**, emulsifiers, stabilisers, thickeners, **MSG (ajinomoto – "Chinese restaurant syndrome", avoid in infants < 12 months)**, anti-caking agents; **must be within FSSAI limits; banned colours – metanil yellow, lead chromate, auramine, rhodamine, Sudan dyes**.
- **Food fortification** – **addition of micronutrients to commonly consumed foods** to improve nutritional quality with minimal risk – **cost-effective public-health strategy**.
- **FSSAI "+F" logo** (Food Safety and Standards (Fortification of Foods) Regulations, 2018) – **5 staples**:
- **Salt – iodine (iodised); double-fortified salt (DFS – iodine + iron)**.
- **Edible oil – vitamins A & D**.
- **Milk – vitamins A & D**.
- **Wheat flour (atta/ maida) – iron, folic acid, vitamin B12** (± zinc, vitamin A, B-complex).
- **Rice – iron, folic acid, vitamin B12** (**fortified rice kernels – FRK, 1 : 100 blending**) – **distributed through PDS, ICDS & PM POSHAN – all government schemes by 2024 (PMGKAY)**.
- **Biofortification** – nutrient-rich crop varieties through breeding (**iron-rich pearl millet (Dhanashakti), zinc wheat/ rice, vitamin A–rich orange maize, golden rice (genetic)**).

- **Supplementation** vs fortification – supplements given as pills/ syrups to target groups (IFA, vitamin A).

8.6 Food Adulteration

- **Adulteration** – **intentional addition/ substitution/ removal** of substances that **lowers quality** or makes food injurious, or **incidental** contamination; **misbranding**.
- **Prevention of Food Adulteration Act (PFA) 1954** → replaced by **Food Safety and Standards Act (FSS Act) 2006** → **FSSAI (Food Safety and Standards Authority of India) – established 2008 (operational 2011), under MoHFW, HQ New Delhi**; **FSSAI licence/ registration number (14 digits)** on packed foods; **penalties – imprisonment & fines (up to life imprisonment if death)**; **Food Safety Officers**, **food testing labs**; **"Eat Right India", "Food Safety on Wheels" (mobile testing vans), DART book (Detect Adulteration with Rapid Test)**; **World Food Safety Day – 7 June**; **World Food Day – 16 October (FAO)**.

Food	Common adulterant	Simple test
Milk	**Water** (most common), starch, urea, detergent, synthetic milk	**Lactometer (specific gravity ~1.028–1.032 – ↓ with water)**; **iodine → blue (starch)**; froth/ lather (detergent); flowing drop test on slanted surface (pure milk leaves white trail)
Ghee/ butter	**Vanaspati, mashed potato/ starch, animal fat**	**Baudouin test (HCl + sugar → crimson red – vanaspati/ sesame oil)**; iodine → blue (starch)
Mustard oil	**Argemone oil**	**Nitric acid test → orange-red/ brown (epidemic dropsy)**
Edible oils	Mineral oil, castor oil	Saponification test (turbidity – mineral oil)
Turmeric powder	**Metanil yellow (dye), lead chromate**, chalk	**Conc. HCl → pink/ violet (metanil yellow)**
Chilli powder	**Brick powder, sawdust, artificial colour (Sudan red)**	**Water – brick powder sinks, sawdust floats, colour streaks**
Black pepper	**Papaya seeds** (dried)	**Alcohol/ water – papaya seeds float**, shrivelled, oval
Coriander/ spices powder	Horse dung, sawdust	Water – floats, smell
Tea leaves	**Used/ exhausted tea leaves (coloured), iron filings**	**Wet filter paper – colour spots**; magnet (iron filings)
Coffee	**Chicory, tamarind seeds, date seeds**	Water – chicory sinks quickly, colours water
Sugar/ jaggery	Chalk powder, washing soda, metanil yellow	Dissolve in water – chalk settles; HCl → effervescence (washing soda)
Honey	Sugar syrup, jaggery	Cotton wick burns (pure); dissolves quickly in water (adulterated)
Pulses (arhar/ toor dal)	**Khesari dal (Lathyrus)**, metanil yellow	Visual (wedge-shaped khesari); HCl → pink
Wheat flour/ rice	Chalk, sand, iron filings, plastic rice	Water; magnet; burning test

Food	Common adulterant	Simple test
Asafoetida (hing)	Soapstone, resin	Burns brightly (pure)
Green vegetables (peas, chillies)	**Malachite green** (colour)	Cotton dipped in liquid paraffin → green
Fruits	**Calcium carbide (artificial ripening – banned)**, wax coating	–
Silver leaf (vark)	Aluminium	Burns to grey ash (aluminium) vs silver ball
Ice cream	Washing powder	Lemon juice → bubbles
Common salt	White powdered stone/ chalk	Water – insoluble chalk

8.7 Food Labelling

- **Mandatory (FSSAI Labelling & Display Regulations, 2020) – name of food, list of ingredients (descending order of weight), nutritional information (energy, protein, CHO, sugar, fat, saturated/ trans fat, sodium per 100 g/ serving), declaration of veg (green dot)/ non-veg (brown/ red dot) logo**, food additives, **name & address of manufacturer, FSSAI logo & licence number**, **net quantity**, **lot/ batch number**, **date of manufacture/ packing**, **"best before"/ "use by"/ expiry date**, country of origin (imports), instructions for use, allergen declaration.
- **"Best before"** – quality; **"use by/ expiry"** – safety (don't consume after).
- **Front-of-pack labelling (proposed)**, **"Indian Nutrition Rating" (star rating)**.
- **Organic – "Jaivik Bharat" logo**; **ISI (BIS – industrial products), AGMARK (agricultural products – honey, spices, ghee)**, **Hallmark (gold)**; **FPO mark (fruit products – earlier)**.

Yaad karne ki trick

- WHO 5 keys – **"Clean, Separate, Cook, Chill, Safe water"**.
- Danger zone – **"5 to 60 °C – bacteria party zone"**.
- Food poisoning fastest – **"Staph Strikes Soon (1–6 hrs)"**; deadliest – **"Botulism – Bottles/ cans – Breathing stops"**.
- Argemone – **"A for Argemone, A for dropsy (Ankle oedema), Acid test (nitric)"**.
- +F foods – **"Salt, Oil, Milk, Atta, Rice" ("SOMAR")**.

PYQ style

- Danger zone of temperature – **5–60 °C**.
- Holder method pasteurisation – **63 °C × 30 min**; HTST – **72 °C × 15–20 s**.
- Test for adequate pasteurisation – **phosphatase test**.
- Milk bacterial quality – **methylene blue reduction test**.
- Father of White Revolution – **Dr Verghese Kurien**.
- Shortest incubation food poisoning – **staphylococcal (1–6 hrs)**.
- Food poisoning with paralysis – **botulism**.
- Fried/ reheated rice – **Bacillus cereus**.
- Lathyrism – **khesari dal (BOAA) – spastic paraplegia**.
- Epidemic dropsy – **argemone oil in mustard oil**; test – **nitric acid**.
- Aflatoxin – **groundnut/ maize – liver cancer**.
- Preservative for fruit juices – **sodium benzoate**; light-coloured fruits – **KMS**.
- Jam preserved by – **high sugar (osmotic)**.
- Irradiated food symbol – **Radura**.
- Fortified rice ratio – **1 : 100 (FRK)**; nutrients – **iron, folic acid, B12**.
- +F oil & milk fortified with – **vitamins A & D**.
- Double-fortified salt – **iodine + iron**.
- FSS Act – **2006**; FSSAI – **2008**; earlier PFA Act – **1954**.
- World Food Safety Day – **7 June**; World Food Day – **16 October**.
- Milk adulteration with water – **lactometer**; starch – **iodine (blue)**.
- Vanaspati in ghee – **Baudouin test**.
- Metanil yellow in turmeric – **HCl → pink**.
- Papaya seeds adulterate – **black pepper**.
- Brick powder adulterates – **chilli powder**.
- Artificial fruit ripening (banned) – **calcium carbide**.
- Veg food label symbol – **green dot**.

9 National Nutrition Programmes & Nutrition Education

9.1 Integrated Child Development Services (ICDS)

- **Launched 2 October 1975** (33 blocks; Ministry of Women & Child Development – MWCD); **world's largest community-based programme for early childhood care**; now under **"Saksham Anganwadi & Poshan 2.0" (2021–26)**.
- **Beneficiaries** – **children 0–6 yrs, pregnant & lactating women, adolescent girls (11–14 yrs out-of-school – Scheme for Adolescent Girls; aspirational districts/ NE)**.
- **Six services** – **1. Supplementary nutrition**, **2. Immunisation**, **3. Health check-ups**, **4. Referral services**, **5. Pre-school (non-formal) education (3–6 yrs)**, **6. Nutrition & health education (women 15–45 yrs)** – immunisation, health check-ups & referral delivered via **health system (ANM/ PHC)**.
- **Anganwadi Centre (AWC)** – **1 per ~400–800 population (rural/ urban), 1 per 150–400 (mini-AWC/ tribal)**; **Anganwadi Worker (AWW) + helper**; **supervisor (~25 AWCs)**, **CDPO** (block); **Poshan Tracker app** (real-time growth monitoring).

Supplementary nutrition norms (Poshan 2.0 – per beneficiary per day, 300 days/yr)

Beneficiary	Energy (kcal)	Protein (g)
Children 6 months–3 yrs (take-home ration – THR)	500	12–15
Children 3–6 yrs (hot cooked meal + morning snack)	500	12–15
Severely underweight/ malnourished children (6 m–6 yrs)	800	20–25
Pregnant & lactating women (THR)	600	18–20
Adolescent girls (SAG)	600	18–20

- **THR** – fortified micronutrient-rich premix (e.g. "**Bal Bhog**", "**Matrushakti**", "**Purna Shakti**" in Gujarat), **millets promoted**.
- **Growth monitoring** – **WHO growth standards** – monthly weighing (< 3 yrs), MCP card; **Village Health, Sanitation & Nutrition Day (VHSND) – monthly at AWC**.

9.2 POSHAN Abhiyaan (National Nutrition Mission)

- **Launched 8 March 2018 (Jhunjhunu, Rajasthan)** – **PM's Overarching Scheme for Holistic Nutrition**; now **Mission Poshan 2.0 (2021)**.
- **Targets (original, by 2022)** – reduce **stunting by 2 % per year**, **undernutrition (underweight) by 2 % p.a.**, **anaemia (children, women, adolescent girls) by 3 % p.a.**, **LBW by 2 % p.a.**; **"Mission 25 by 2022"** – reduce stunting from 38.4 % to 25 %.
- **Pillars** – ICT-enabled real-time monitoring (**Poshan Tracker**), convergence of schemes, **behaviour change – Jan Andolan**, incentives, **Community-Based Events (CBEs – "Godh Bharai", "Annaprashan")**.
- **Poshan Maah – September (every year) (Rashtriya Poshan Maah)**; **Poshan Pakhwada – March**; **National Nutrition Week – 1–7 September**.

- **First 1000 days (conception to 2 yrs)** – critical window of opportunity.

9.3 PM POSHAN (Mid-Day Meal Scheme)

- **Started 15 August 1995** as **National Programme of Nutritional Support to Primary Education (NP-NSPE)**; **cooked mid-day meal from 2004 (Supreme Court order 2001)**; **renamed PM POSHAN (Pradhan Mantri Poshan Shakti Nirman) – 2021**; Ministry of Education; **Tamil Nadu – pioneer (1956/ 1982 – K. Kamaraj/ MGR)**.
- **Coverage** – **Bal Vatika (pre-primary) & classes 1–8** in government & aided schools.
- **Norms per child per day** – **primary (classes 1–5): 450 kcal & 12 g protein**; **upper primary (6–8): 700 kcal & 20 g protein**; food grains – 100 g (primary)/ 150 g (upper primary); **micronutrients (iron, folic acid, vitamin A), deworming**; **fortified rice**; **tithi bhojan** (community contribution), **school nutrition gardens**.
- **Objectives** – improve nutritional status, **↑ enrolment, attendance & retention**, socialisation (caste barriers), relief in drought areas.

9.4 Micronutrient Programmes

Programme	Key points
National Vitamin A Prophylaxis Programme (1970)	**9 doses** – **1 lakh IU at 9 months** (with MR-1), **2 lakh IU at 16–18 months**, then **2 lakh IU every 6 months up to 5 yrs**; biannual rounds (**"Bal Swasthya Poshan Maah"** in some states); treatment of xerophthalmia/ measles/ SAM
Anaemia Mukt Bharat (2018) (earlier National Nutritional Anaemia Prophylaxis Programme – 1970, NIPI 2013)	**6×6×6 strategy**; IFA – **biweekly syrup (6–59 m)**, **pink weekly (5–9 yrs)**, **blue weekly (10–19 yrs – WIFS)**, **red weekly (WRA)**, **red daily × 180 days (pregnancy & lactation)**; **deworming (NDD – 10 Feb & 10 Aug)**; test & treat; fortified foods; target ↓ anaemia 3 % points/yr
NIDDCP (1992; NGCP 1962)	**Universal salt iodisation – ≥ 30 ppm production, ≥ 15 ppm consumer**; **ban on non-iodised salt (2006)**; IDD surveys, urinary iodine lab, health education; **Global IDD Prevention Day – 21 October**
Zinc + ORS	Diarrhoea management (IDCF – Intensified Diarrhoea Control Fortnight / **STOP Diarrhoea Campaign**, June–July)
Calcium & vitamin D supplementation	Pregnant & lactating women (MoHFW)
Folic acid	Preconception & pregnancy
Food fortification	**+F (salt, oil, milk, atta, rice) – fortified rice in PDS/ ICDS/ PM POSHAN**
National Fluorosis Programme (NPPCF)	Safe water, defluoridation, nutrition (calcium, vitamin C)

9.5 Malnutrition Management & Maternal-Child Schemes

- **Nutrition Rehabilitation Centres (NRCs – NHM, 2008)** – facility-based care of **complicated SAM** (14 days stay, mother's wage compensation, counselling, follow-up); **Community-based Management of Acute Malnutrition (CMAM)**.

- **MAA (Mothers' Absolute Affection, 2016)** – promotion of breastfeeding (early initiation, EBF, IYCF counselling by ASHA/ ANM).
- **Home-Based Newborn Care (HBNC) & Home-Based Care for Young Child (HBYC – 3, 6, 9, 12, 15 months visits – ASHA)** – feeding counselling, growth monitoring.
- **PMMVY (2017)** – **₹5000** conditional cash for first child (₹6000 for second child if girl – PMMVY 2.0); **Janani Suraksha Yojana (2005)** – institutional delivery cash.
- **RBSK (Rashtriya Bal Swasthya Karyakram, 2013)** – screening for **4 Ds: Defects at birth, Diseases, Deficiencies, Development delays** (0–18 yrs).
- **RKSK (2014)** – adolescent health including nutrition (WIFS, counselling).
- **Mission Indradhanush** – immunisation (measles → malnutrition link).
- **LaQshya, SUMAN** – maternal care.
- **Gujarat initiatives** – **"Kuposhan Mukt Gujarat Mahabhiyan", "Mission Balam Sukham" (SAM/ MAM management – Child Malnutrition Treatment Centres – CMTC & Child Development Nutrition Centres)**, **"Purna Yojana" (adolescent girls)**, **"Doodh Sanjivani Yojana"** (fortified milk in schools/ AWCs – tribal), **"Poshan Sudha"** (hot meals to pregnant/ lactating women – tribal), **"Matrushakti, Bal Shakti, Purna Shakti"** (THR), **"Mukhyamantri Matrushakti Yojana"** (2022 – 1000 days nutrition kit).

9.6 Food Security Programmes

- **National Food Security Act (NFSA), 2013** – legal entitlement – **up to 75 % rural & 50 % urban population**; **5 kg food grains per person per month at ₹3/2/1 (rice/ wheat/ coarse grains)** – **Antyodaya Anna Yojana (AAY – poorest of poor) – 35 kg per household per month**; **maternity benefit ≥ ₹6000**; ICDS & MDM entitlements; **eldest woman (≥ 18 yrs) – head of household for ration card**.
- **PMGKAY (Pradhan Mantri Garib Kalyan Anna Yojana, 2020)** – **free food grains** (merged with NFSA – free from 1 January 2023, extended 5 yrs till 2028).
- **Targeted Public Distribution System (TPDS, 1997)**; **One Nation One Ration Card (2019)**.
- **Annapurna scheme (2000)** – 10 kg free food grains for destitute elderly not getting pension.
- **National Food Security Mission (agriculture – production)**.
- **Food for Work, Antyodaya**.

9.7 Global Goals & Organisations

- **SDG 2 – "Zero Hunger" (end hunger, achieve food security, improved nutrition, sustainable agriculture by 2030)**; SDG 3 (health).
- **WHO Global Nutrition Targets 2025 (World Health Assembly 2012)** – **40 % ↓ in stunting (under-5)**, **50 % ↓ anaemia in women of reproductive age**, **30 % ↓ LBW**, **no ↑ in childhood overweight**, **↑ EBF to ≥ 50 %**, **wasting < 5 %** (extended to 2030).
- **UN Decade of Action on Nutrition (2016–2025)**; **SUN (Scaling Up Nutrition) movement**.
- **Global Hunger Index** (India ranks low; 2023 – 111/125, 2024 – 105/127); **State of Food Security & Nutrition in the World (SOFI – FAO)**.
- **Organisations** – **FAO (Food & Agriculture Organisation, 1945 – Rome; World Food Day 16 Oct)**, **WFP (World Food Programme – Nobel Peace Prize 2020)**, **UNICEF (1946)**, **WHO**, **CARE (Cooperative for Assistance & Relief Everywhere – supplementary feeding)**, **ICMR-NIN (Hyderabad)**, **CFTRI (Central Food Technological Research Institute – Mysuru)**, **National Nutrition Monitoring**

Bureau (NNMB, 1972 – NIN), **Food & Nutrition Board (MWCD)**, **Nutrition Foundation of India (C. Gopalan, Delhi)**, **IIPS Mumbai – NFHS**; **Comprehensive National Nutrition Survey (CNNS 2016–18)** – first national survey of micronutrients in children & adolescents.

- **NFHS-5 (2019–21) key data** – stunting **35.5 %**, wasting **19.3 %**, severe wasting **7.7 %**, underweight **32.1 %**, overweight (U5) **3.4 %**; anaemia – children (6–59 m) **67.1 %**, women **57 %**, pregnant women **52.2 %**, men **25 %**; **EBF (< 6 months) 63.7 %**; early initiation of breastfeeding **41.8 %**; minimum acceptable diet (6–23 m) **11.3 %**; women BMI < 18.5 – **18.7 %**; women overweight/ obese – **24 %**.

9.8 Nutrition Education & Nurse's Role

- **Nutrition education** – process of helping people **adopt healthy food habits** using available resources – **change knowledge, attitude & practice (KAP)**.
- **Target groups (priority/ vulnerable)** – **pregnant & lactating women, infants & preschool children (most vulnerable), adolescent girls, elderly, sick & convalescent, low-income families**.
- **Methods** – **individual** (counselling at clinic/ home visits – most effective for behaviour change), **group** (**demonstration – most effective method for teaching cooking/ recipes**, group discussion, role play, **mothers' meetings at AWC, VHSND**), **mass** (radio, TV, posters, exhibitions, social media, Poshan Maah campaigns, **street plays/ nukkad natak**, puppet shows).
- **Audio-visual aids** – **flash cards, flannel graph, charts, posters, models (food models), real foods, videos**.
- **Principles** – start from known to unknown, simple language, local & low-cost foods (**"seasonal, local, affordable"**), cultural acceptability, involve family decision-makers (mother-in-law/ husband), repeat & reinforce, practical demonstration, **dispel food fads & taboos**, evaluation.
- **Key messages** – EBF 6 months, timely complementary feeding, IFA compliance, iodised salt, hand washing, diverse diet (GLV, pulses, milk, eggs), kitchen gardens, safe water, growth monitoring, de-worming, family planning (spacing).
- **Nurse's role** – **nutritional assessment & screening (MUAC, weight, Hb)**, **growth monitoring (MCP card)**, identify & refer malnourished (NRC), **IFA/ vitamin A/ deworming distribution & compliance**, breastfeeding & IYCF counselling, **cooking demonstrations**, diet counselling in diseases, coordination with AWW/ ASHA (convergence – "**AAA – ASHA, AWW, ANM**"), record keeping & reporting, school health, community surveys, health education campaigns, research.

♫ Yaad karne ki trick

- ICDS 6 services – **"SIH-RPN"** – Supplementary nutrition, Immunisation, Health check-up, Referral, Pre-school education, Nutrition & health education.
- ICDS – **"500-600-800"** – children 500, pregnant/ lactating 600, severely malnourished 800 kcal.
- MDM – **"450-12 / 700-20"** – primary 450 kcal & 12 g protein; upper primary 700 kcal & 20 g.
- POSHAN Abhiyaan targets – **"2-2-3-2"** – stunting 2, underweight 2, anaemia 3, LBW 2 (% per year).
- Poshan Maah – **September**; Pakhwada – **March**.

PYQ style

- ICDS launched – **2 October 1975**; ministry – **MWCD**.
- ICDS services – **6**; worker – **Anganwadi worker**.
- Supplementary nutrition for children (6 m–6 yrs) – **500 kcal, 12–15 g protein**.
- Pregnant & lactating women – **600 kcal, 18–20 g protein**.
- Severely malnourished children – **800 kcal, 20–25 g protein**.
- POSHAN Abhiyaan launched – **8 March 2018, Jhunjhunu (Rajasthan)**.
- Poshan Maah – **September**; National Nutrition Week – **1–7 September**.
- Mid-day meal started – **1995**; renamed – **PM POSHAN (2021)**.
- MDM primary norm – **450 kcal & 12 g protein**; upper primary – **700 kcal & 20 g**.
- Vitamin A prophylaxis programme – **1970**.
- Anaemia Mukt Bharat – **2018**; strategy – **6×6×6**.
- NIDDCP – **1992**.
- NRC – **complicated SAM**.
- MAA programme – **breastfeeding promotion (2016)**.
- NFSA – **2013**; grains per person – **5 kg/month**; AAY – **35 kg/household**.
- First 1000 days – **conception to 2 years**.
- SDG 2 – **Zero Hunger**.
- WHO global target – stunting ↓ **40 %**, anaemia in women ↓ **50 %**, LBW ↓ **30 %**.
- NNMB established by – **NIN (1972)**.
- CFTRI located at – **Mysuru**.
- NFHS-5 stunting – **35.5 %**; anaemia in children – **67.1 %**.
- Most vulnerable group for malnutrition – **preschool children/ infants**.
- Best method to teach cooking – **demonstration**.
- Most effective for behaviour change – **individual counselling**.
- Mission Balam Sukham – **Gujarat (child malnutrition)**.

10 Quick Revision – Master Tables & 150 One-Liners

10.1 Richest / Best Sources

Nutrient	Richest / best source
Protein (plant)	**Soybean (~40 %)**
Protein quality (reference)	**Egg**
Calcium	**Milk (best absorbed)**; ragi (cereal); sesame seeds
Iron (plant)	**Garden cress seeds (halim), GLV, jaggery, rice flakes**
Iron (animal)	**Liver, meat (haem iron)**
Vitamin A	**Fish liver oil**; plant – **carrot, GLV (carotene)**
Vitamin D	**Sunlight**; dietary – fish liver oil
Vitamin E	**Wheat germ oil**
Vitamin K	**Green leafy vegetables**
Vitamin C	**Amla**; then guava
Thiamine (B1)	**Whole grains, rice bran, yeast**
Riboflavin (B2)	**Milk**
Niacin (B3)	**Groundnut** (plant), meat
Folic acid	**GLV, liver**
Vitamin B12	**Liver** (only animal foods)
Iodine	**Sea food; iodised salt**
Zinc	**Oysters/ sea food, meat**
Fluoride	**Drinking water, tea**
ω-3 fatty acids	**Fish oil, flaxseed**
Saturated fat (oil)	**Coconut oil**
Fibre	**Whole grains, bran, isabgol, vegetables**
Potassium	**Banana, coconut water, orange juice**
Energy (most concentrated)	**Fat (9 kcal/g)**

10.2 Deficiency Diseases

Deficiency	Disease / sign
Protein-energy	**Kwashiorkor, marasmus**
Vitamin A	**Night blindness, Bitot's spots, xerophthalmia, keratomalacia**
Vitamin D	**Rickets, osteomalacia**

Deficiency	Disease / sign
Vitamin E	**Haemolytic anaemia (preterm)**
Vitamin K	**Haemorrhagic disease of newborn**
B1 (thiamine)	**Beriberi, Wernicke-Korsakoff**
B2 (riboflavin)	**Angular stomatitis, cheilosis, magenta tongue**
B3 (niacin)	**Pellagra (3 Ds)**
B6 (pyridoxine)	**Convulsions (infants), neuropathy**
B9 (folate)	**Megaloblastic anaemia, NTD**
B12	**Pernicious anaemia, SACD**
Vitamin C	**Scurvy**
Iron	**Microcytic hypochromic anaemia, koilonychia**
Iodine	**Goitre, cretinism**
Calcium	**Tetany, osteoporosis, rickets**
Zinc	**Growth retardation, hypogeusia, acrodermatitis enteropathica**
Fluoride (low)	**Dental caries**
Selenium	**Keshan disease**
EFA	**Phrynoderma**
Copper	**Anaemia, Menkes**

10.3 RDA Quick Table (ICMR-NIN 2020 – approx.)

Nutrient	Adult man	Adult woman	Pregnancy	Lactation
Energy (moderate)	~2710 kcal	~2130 kcal	**+350**	**+600 (0–6 m)**
Protein	~54 g	~46 g	**+9.5/ +22 g**	**+17 g**
Calcium	1000 mg	1000 mg	1000 mg	1000 mg
Iron	**19 mg**	**29 mg**	27 mg	23 mg
Folate	~300 µg	~220 µg	**~570 µg**	~330 µg
Vitamin C	**80 mg**	**65 mg**	+20	+40
Iodine	**150 µg**	150 µg	**250 µg**	**250 µg**
Vitamin D	600 IU	600 IU	600 IU	600 IU

10.4 150 Rapid-Fire One-Liners

150 one-liners

1. Father of nutrition – **Lavoisier**.
2. NIN – **Hyderabad**.
3. Most essential nutrient – **water**.
4. Carbohydrate energy – **4 kcal/g**.
5. Protein energy – **4 kcal/g**.
6. Fat energy – **9 kcal/g**.
7. Alcohol energy – **7 kcal/g**.
8. 1 kcal – **4.184 kJ**.
9. Largest energy expenditure – **BMR**.
10. BMR highest per kg – **infants**.
11. BMR ↑ per °C fever – **13 %**.
12. Highest SDA – **protein**.
13. RDA covers – **97.5 %**.
14. Reference man (2020) – **65 kg**.
15. Reference woman – **55 kg**.
16. Pregnancy extra energy – **350 kcal**.
17. Lactation extra energy – **600 kcal**.
18. Protein adult – **0.83 g/kg**.
19. Salt limit – **< 5 g/day**.
20. Cereal limiting amino acid – **lysine**.
21. Pulse limiting amino acid – **methionine**.
22. Sweetest sugar – **fructose**.
23. Milk sugar – **lactose**.
24. Animal starch – **glycogen**.
25. Glucose GI – **100**.
26. Low GI – **≤ 55**.
27. Soluble fibre – **↓ cholesterol**.
28. Fibre requirement – **25–40 g**.
29. Essential amino acids – **9**.
30. Reference protein – **egg**.
31. Best protein quality index – **NPU**.
32. Highest plant protein – **soybean**.
33. Essential fatty acids – **linoleic, α-linolenic**.
34. EFA deficiency – **phrynoderma**.
35. Highest SFA oil – **coconut**.
36. Most harmful fat – **trans fat**.
37. FSSAI trans fat limit – **2 %**.
38. Cholesterol only in – **animal foods**.
39. Good cholesterol – **HDL**.
40. Desirable cholesterol – **< 200 mg/dl**.

41. MCT used in – **malabsorption**.
42. Body water – **60 %**.
43. ORS osmolarity – **245 mOsm/L**.
44. Zinc in diarrhoea – **20 mg × 14 days**.
45. Vitamine term – **Casimir Funk**.
46. Fat-soluble vitamins – **ADEK**.
47. Night blindness – **vitamin A**.
48. Bitot's spots – **vitamin A**.
49. Vitamin A 1st dose – **1 lakh IU, 9 months**.
50. Vitamin A total doses – **9**.
51. Vitamin A toxicity infant – **bulging fontanelle**.
52. Sunshine vitamin – **D**.
53. Active vitamin D – **calcitriol**.
54. Earliest rickets sign – **craniotabes**.
55. Rickety rosary – **vitamin D**.
56. Infant vitamin D – **400 IU**.
57. Antioxidant (fat-soluble) – **vitamin E**.
58. Newborn vitamin K – **1 mg IM**.
59. Beriberi – **B1**.
60. Wernicke – **B1 (thiamine)**.
61. Angular stomatitis – **B2**.
62. Pellagra – **niacin**.
63. Pellagra – **maize/ jowar eaters**.
64. INH neuropathy – **B6**.
65. Avidin – **biotin**.
66. NTD prevention – **folic acid 400 µg**.
67. B12 only in – **animal foods**.
68. B12 absorbed – **terminal ileum**.
69. Scurvy – **vitamin C**.
70. Richest vitamin C – **amla**.
71. Most labile vitamin – **C**.
72. Infantile scurvy – **Barlow's disease**.
73. Most abundant mineral – **calcium**.
74. Calcium RDA – **1000 mg**.
75. Richest cereal calcium – **ragi**.
76. Tetany – **hypocalcaemia**.
77. Iron storage – **ferritin**.
78. Iron transport – **transferrin**.
79. Iron absorption site – **duodenum**.
80. Iron enhancer – **vitamin C**.
81. Iron inhibitor – **tea, phytates**.
82. Iron RDA woman – **29 mg**.
83. Earliest iron depletion marker – **serum ferritin**.

84. Koilonychia – **iron deficiency**.
85. Pregnancy anaemia – **Hb < 11**.
86. Severe anaemia – **Hb < 7**.
87. IFA pregnancy – **60 mg + 500 µg, 180 days**.
88. WIFS tablet colour – **blue**.
89. Deworming days – **10 Feb & 10 Aug**.
90. Iodine RDA – **150 µg**.
91. Pregnancy iodine – **250 µg**.
92. Iodised salt consumer – **15 ppm**.
93. Goitrogens – **cabbage, cauliflower**.
94. Preventable mental retardation – **iodine deficiency**.
95. Fluoride optimum – **0.5–0.8 ppm**.
96. Dental fluorosis – **> 1.5 ppm**.
97. Defluoridation – **Nalgonda technique**.
98. Keshan disease – **selenium**.
99. Wilson's disease – **copper**.
100. Weight gain pregnancy – **11.5–16 kg**.
101. Breastfeeding initiation – **within 1 hr**.
102. EBF – **6 months**.
103. Continue BF – **2 yrs**.
104. Colostrum – **IgA, protein, vitamin A**.
105. Highest lactose milk – **human**.
106. Highest fat milk – **buffalo**.
107. Milk production – **prolactin**.
108. Milk ejection – **oxytocin**.
109. Hindmilk – **fat-rich**.
110. Good attachment – **lower lip everted**.
111. EBM fridge – **24 hrs**.
112. Complementary feeding – **6 months**.
113. Energy density – **add oil/ ghee**.
114. Honey < 1 yr – **botulism**.
115. Bottle feeding hazard – **diarrhoea**.
116. Kwashiorkor term – **Cicely Williams**.
117. Kwashiorkor hallmark – **oedema**.
118. Flag sign – **kwashiorkor**.
119. Marasmus – **no oedema, old-man face**.
120. SAM MUAC – **< 11.5 cm**.
121. SAM WFH – **< −3 SD**.
122. MAM MUAC – **11.5–12.5 cm**.
123. Complicated SAM – **NRC**.
124. SAM ORS – **ReSoMal**.
125. SAM starter feed – **F-75**.
126. Refeeding syndrome – **hypophosphataemia**.

127. Stunting – **low height-for-age**.
128. Wasting – **low weight-for-height**.
129. Gomez – **weight-for-age**.
130. Indian obese BMI – **≥ 25**.
131. Indian waist (women) – **≥ 80 cm**.
132. Skinfold – **Harpenden calipers**.
133. Commonest diet survey – **24-hr recall**.
134. DASH diet – **hypertension**.
135. CKD diet – **low protein, K, P, Na**.
136. Coeliac – **gluten-free**.
137. Burns – **highest calorie & protein**.
138. Gout – **low purine**.
139. PKU – **low phenylalanine**.
140. Preferred nutrition route – **enteral**.
141. Tube feed position – **≥ 30–45° head up**.
142. TPN complication – **infection, hyperglycaemia**.
143. Danger zone – **5–60 °C**.
144. Pasteurisation test – **phosphatase**.
145. Epidemic dropsy – **argemone oil**.
146. Lathyrism – **khesari dal**.
147. FSS Act – **2006**.
148. ICDS – **1975**.
149. POSHAN Abhiyaan – **8 March 2018**.
150. Poshan Maah – **September**.



Exam strategy – Asha Didi ki last advice

- GSSSB Staff Nurse paper me **Nutrition se ~5–8 direct questions** aate hain, aur Community Health, OBG aur Paediatrics ke questions (breastfeeding, IFA, vitamin A, SAM, ICDS) bhi isi par based hote hain.
- Sabse zyada pooche jaate hain – **vitamin deficiency diseases, richest sources, kcal values, RDA (pregnancy/ lactation), breastfeeding & complementary feeding, anaemia & IFA, SAM/ MUAC, ICDS & POSHAN, iodine/ fluoride numbers, therapeutic diets, food adulteration tests**.
- **Numbers ki list** banao aur roz revise karo – 4-4-9-7 kcal, +350/+600 kcal, 60 mg + 500 µg IFA, 1 lakh/ 2 lakh IU, 11.5 cm MUAC, 15/30 ppm iodine, 0.5–0.8 ppm fluoride, 500/600/800 kcal ICDS.
- MCQ Bank PDF ke saare questions **2 baar** solve karo aur galat hue questions ko notebook me likho.

— Nutrition · GSSSB Staff Nurse Preparation · Padhte raho, revise karte raho! —